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Power tool attachment

Abstract

A power tool attachment is configured to connect to a power tool at a static interface and a dynamic interface. The static interface statically fixes the power tool and power tool attachment together. The dynamic interface provides rotational motion generated by the power tool to the power tool attachment. The rotational motion is provided to a drive of the power tool attachment that converts the rotational motion to a linear reciprocating output.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application claims priority to U.S. Provisional Application No. 63/466,930 filed May 16, 2023 and entitled “POWER TOOL ATTACHMENT,” and claims priority to U.S. Provisional Application No. 63/601,782 filed Nov. 22, 2023 and entitled “POWER TOOL ATTACHMENT,” and claims priority to U.S. Provisional Application No. 63/555,273 filed Feb. 19, 2024 and entitled “POWER TOOL ATTACHMENT,” and claims priority to U.S. Provisional Application No. 63/574,378 filed Apr. 4, 2024 and entitled “POWER TOOL ATTACHMENT,” the disclosures of which is hereby incorporated by reference in their entireties.

BACKGROUND

(1) This disclosure relates generally to power tool attachments. More specifically, this disclosure relates to a power tool appliance configured for airless spraying of paints and other fluids.

(2) Power tools, such as impact drivers, drills, or other motorized tools output rotational motion. Such power tools can be capable of drilling in screws or other fasteners, particularly when hex bits are installed in the power tool for interfacing with the fastener (e.g., a flathead or Phillips head hex bits), the power tool rotating the hex bits to rotate the fasteners. Such power tools are typically battery-powered, but in some versions can include a cord for receiving power via electrical outlet. Such power tools typically include an electric motor, a handle, a trigger, and a power source. Power tool appliances can connect to the power tool to receive the rotational output from the power tool to operate the power tool appliance.

SUMMARY

(3) According to an aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; and a mount connected to the housing, the mount configured to interface with the power tool to statically connect the power tool to the housing.

(4) According to an additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a mount connected to the housing, the mount configured to interface with the power tool to statically connect the power tool to the housing; and a shaft aligner supported on the housing, the shaft aligner configured to bias the housing vertically upwards relative to the power tool.

(5) According to another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a support assembly connected to the housing and extending from the housing, the support assembly configured to interface with the power tool to statically connect the power tool to the

housing; and a spring disposed between the housing and the support assembly, the spring configured to bias the housing upwards and away from the support assembly to align the power tool and the power tool assembly.

(6) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a support assembly connected to the housing and extending from the housing, the support assembly configured to interface with the power tool to statically connect the power tool to the housing; and a first plurality of springs configured to bias the housing upwards and away from the support assembly to align the power tool and the power tool assembly.

(7) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a support assembly connected to the housing and extending from the housing, the support assembly configured to interface with the power tool to statically connect the power tool to the housing; a first spring configured to bias the support assembly longitudinally rearward away from the housing and vertically upwards to align the power tool and the power tool assembly; and a tensioner configured to bias a mount of the support assembly to align the power tool and the power tool assembly.

(8) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool. The power tool appliance is configured to preload the power tool.

(9) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; an arm extending from the housing, the arm configured to connect the power tool to the housing; and a strut configured to engage the arm to orient the arm relative to the housing.

(10) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; and an arm connected to the housing and configured to statically connect the power tool to the housing.

(11) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; and an arm connected to the housing and configured to statically connect the power tool to the housing, wherein the arm is reconfigurable between a contracted state and an extended state to vary

a length of the arm.

(12) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; an input shaft extending out a rear end of the housing, the input shaft configured to interface with the power tool to receive the rotational input; and an arm movably connected to the housing at an arm pivot, the arm configured to statically connect the power tool to the housing.

(13) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive including a drive shaft configured to rotate on a drive axis; a rotational input shaft projecting rearward relative to a rear side of the housing, the rotational input shaft configured to dynamically connect the drive to the power tool to provide the rotational input from the power tool to the drive, wherein the rotational input shaft is unitary with the drive shaft.

(14) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive including a drive shaft configured to rotate on a drive axis; a rotational input shaft projecting rearward relative to a rear side of the housing, the rotational input shaft configured to dynamically connect the drive to the power tool to provide the rotational input from the power tool to the drive, wherein the rotational input shaft includes a keyed exterior.

(15) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive including a drive shaft configured to rotate on a drive axis; a rotational input shaft projecting rearward relative to a rear side of the housing, the rotational input shaft configured to dynamically connect the drive to the power tool to provide the rotational input from the power tool to the drive, wherein the rotational input shaft and a portion of the drive configured to exert axial force to cause axial movement of an output of the drive are formed from a single piece of metal.

(16) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive including a drive shaft configured to rotate on a drive axis; a rotational input shaft projecting rearward relative to a rear side of the housing, the rotational input shaft configured to dynamically connect the drive to the power tool to provide the rotational input from the power tool to the drive, wherein the rotational input shaft is anchored in the housing and is connectable to the power tool to receive the rotational input, wherein the rotational input shaft is not part of the power tool.

(17) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a pump at least partially disposed within the housing, the pump connected to the drive to receive the linear reciprocating output from the drive, wherein the pump including a pump inlet that

is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump; and a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray. The power tool appliance is reconfigurable between a suction feed state in which the pump is configured to draw the spray fluid upward into the pump and a gravity feed state in which the spray fluid is configured to flow downward into the pump.

(18) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a pump at least partially disposed within the housing, the pump connected to the drive to receive the linear reciprocating output from the drive, wherein the pump including a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump; a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray; and an arm connected to the housing and movable relative to the housing, the arm configured to statically connect the power tool to the housing. The housing is configured to pivot relative to the arm between a suction feed state, in which the pump inlet is oriented downward such that the pump is configured to draw the spray fluid upward into the pump, and a gravity feed state, in which the pump inlet is oriented upward such that the spray fluid is configured to flow downward into the pump.

(19) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a pump at least partially disposed within the housing, the pump connected to the drive to receive the linear reciprocating output from the drive, wherein the pump including a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump; and a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray. The housing is configured to flip between a suction feed state, in which the pump inlet is oriented downward such that the pump is configured to draw the spray fluid upward into the pump, and a gravity feed state, in which the pump inlet is oriented upward such that the spray fluid is configured to flow downward into the pump.

(20) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a pump at least partially disposed within the housing, the pump connected to the drive to receive the linear reciprocating output from the drive, wherein the pump including a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump; a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray; an arm connected to the housing and movable relative to the housing; and a mount connected to the arm and movable relative to the arm. The mount and the arm are configured to statically connect the power tool to the housing. The housing is configured to pivot relative to the arm between a suction feed state, in which the pump inlet is oriented downward such that the pump is configured to draw the spray fluid upward into the pump, and a gravity feed state, in which the pump inlet is oriented upward such that the spray fluid is configured to flow downward into the pump.

(21) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a mount connected to the housing, the mount configured to interface with the power tool to statically connect the power tool to the housing; and a fixation configured to wrap around the mount and a base of the power tool to fix the power tool to the mount.

(22) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a mount connected to the housing, the mount configured to interface with the power tool to statically connect the power tool to the housing; and a fixation configured to extend through a slot that extends laterally through the mount to wrap around the mount and a base of the power tool to fix the power tool to the mount.

(23) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance is configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool. The power tool appliance includes a housing; a turbine at least partially disposed within the housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool and configured to provide rotational motion to the turbine to drive the turbine; a duct disposed downstream of the turbine, the duct narrowing from the turbine and configured to direct airflow to a spray orifice; and a mount connected to the housing, the mount configured to interface with the power tool to statically connect the power tool to the housing.

(24) According to yet another additional or alternative aspect of the present disclosure, a method includes inserting an input shaft extending out from a housing of a power tool appliance into a bit receiver of a power tool; mounting the power tool to a mount of the power tool appliance to statically connect the power tool relative to the housing; and pulling the trigger of the power tool to cause an output of the power tool appliance.

(25) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance including a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a pump at least partially disposed within the housing, the pump connected to the drive to receive a linear reciprocating output from the drive, wherein the pump includes an inlet that is exposed through the housing, wherein the inlet is configured to receive spray fluid into the pump; a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray; a support assembly connected to the housing and extending from the housing, the support assembly configured to interface with the power tool to statically connect the power tool to the housing; and a biaser configured to bias the support assembly to align the power tool and the drive.

(26) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance including a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power

tool; a pump at least partially disposed within the housing, the pump connected to the drive to receive a linear reciprocating output from the drive, wherein the pump includes an inlet that is exposed through the housing, wherein the inlet is configured to receive spray fluid into the pump; a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray; a support assembly including an arm connected to the housing and extending from the housing, the support assembly configured to interface with the power tool to statically connect the power tool to the housing; and a spring disposed between the arm and the housing, the spring configured to bias the housing upward and away from the support assembly to align the power tool and the drive.

(27) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance including a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a pump at least partially disposed within the housing, the pump connected to the drive to receive a linear reciprocating output from the drive, wherein the pump includes an inlet that is exposed through the housing, wherein the inlet is configured to receive spray fluid into the pump; a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray; a support assembly including an arm connected to the housing and extending from the housing, the support assembly configured to interface with the power tool to statically connect the power tool to the housing; and a spring disposed within the arm, the spring configured to adjust a length of the arm and configured to bias the arm towards a contracted state to align the power tool and the drive.

(28) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance including a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool, the drive configured to rotate on a drive axis; a pump at least partially disposed within the housing, the pump connected to the drive to receive a linear reciprocating output from the drive, wherein the pump includes an inlet that is exposed through the housing, wherein the inlet is configured to receive spray fluid into the pump; a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray; and a support assembly connected to the housing and extending from the housing, the support assembly configured to interface with the power tool to statically connect the power tool to the housing; wherein a piston of the pump is vertically aligned with the drive and radially offset from the drive axis.

(29) According to yet another additional or alternative aspect of the present disclosure, a power tool appliance for use with a power tool that outputs rotational motion for one or both of drilling and fastening, the power tool appliance configured to attached to, and receive rotational input from, the power tool, the power tool appliance including a housing; a drive at least partially located within the housing, the drive comprising an input shaft having an end that is configured to be received by the power tool, the input shaft extending into the housing, the drive converting rotational motion received from the power tool through the input shaft into a reciprocating motion; and a support connected to the housing, the support configured to interface with the power tool to statically connect the power tool to the housing, the support comprising an arm that extends away from the housing and that connects between the power tool and the housing.

(30) According to yet another additional or alternative aspect of the present disclosure, a method of using a power tool appliance to convert rotational motion output from a power tool to reciprocating motion, the method including mounting the power tool to the power tool appliance, the power tool

appliance comprising: a housing; a drive at least partially located within the housing, the drive comprising an input shaft, the input shaft extending out from the housing; and a support connected to the housing, the support configured to interface with the power tool to statically connect the power tool to the housing, the support comprising an arm that extends away from the housing and which connects between the power tool and the housing, wherein mounting comprises the input shaft being received within the power tool; and operating the power tool appliance via the power tool by the drive converting rotational motion received from the power tool through the input shaft into a reciprocating motion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram of a tool system.
- (2) FIG. 2A is a first side elevational view of a tool system.
- (3) FIG. 2B is a second side elevational view of the tool system.
- (4) FIG. 3 is an isometric view of power tool appliance.
- (5) FIG. 4A is an isometric cross-sectional view taken along line 4-4 in FIG. 3.
- (6) FIG. 4B is a cross-sectional view taken along line 4-4 in FIG. 3.
- (7) FIG. 5 is a side elevational view of the power tool system showing preloading of the power tool.
- (8) FIG. 6 is an isometric view of the power tool appliance in a suction feed state.
- (9) FIG. 7A is an isometric view of the power tool appliance in a gravity feed state.
- (10) FIG. 7B is an isometric view of the power tool appliance in the gravity feed state and showing a reservoir connected to the power tool appliance.
- (11) FIG. 8 is an isometric view of a drive.
- (12) FIG. 9 is a cross-sectional view taken along line 9-9 in FIG. 8.
- (13) FIG. 10A is a first isometric view of a shaft aligner.
- (14) FIG. 10B is a second isometric view of the shaft aligner.
- (15) FIG. 11A is an isometric view of a support assembly of the power tool attachment in a contracted state.
- (16) FIG. 11B is an isometric view of the support assembly in an extended state.
- (17) FIG. 12 is an isometric view of a power tool appliance with an alternative support assembly.
- (18) FIG. 13 is an isometric view of a power tool appliance with an alternative support assembly.
- (19) FIG. 14 is a partially schematic side view of a tool system.

DETAILED DESCRIPTION

- (20) Power tool appliances according to the present disclosure can be connected to a variety of power tools. Such power tools can be impact drivers, drills, or other motorized tools which output rotational motion. Such power tools can be capable of drilling in screws or other fasteners, particularly when hex bits are installed in the power tool for interfacing with the fastener (e.g., a flathead or Phillips head hex bits, among others), the power tool rotating the hex bits to rotate the fasteners. Such power tools are typically battery-powered, but in some versions can include a cord for receiving power via an electrical outlet. Such power tools typically include an electric motor, a handle, a trigger, and a power source. The term “power tool” can refer to all types of power tool options.
- (21) The power tool appliance interfaces with the power tool to receive rotational output from the power tool so that the power tool appliance can perform an operation. For example, the power tool appliance may receive rotational motion from the power tool and convert the rotational motion into linear reciprocating motion, such as for operating a pump or a saw, amongst other options. In various examples, the power tool appliance receives the rotational motion input from the power

tool and does not convert the rotational motion into linear reciprocating motion, but instead the operation within the power tool uses the rotational motion. For example, the rotational motion may be used to turn a turbine to generate airflow, such as for high volume, low pressure (HVLP) spraying of paints, stains, finishes, and/or other coatings.

(22) The main example of a power tool appliance disclosed herein is that of a sprayer, such as for spraying paints, stains, finishes, and/or other coatings. It is understood, however, that the teachings of the present disclosure apply to other types of power tool appliances which are not sprayers. As such, while a sprayer is used as the main example, the features can be embodied in other power tool appliances that perform functions other than spraying.

(23) A power tool appliance according to aspects of the disclosure can be configured to connect to a power tool to receive a dynamic driving input from the power tool. The power tool is statically connected to the power tool appliance to form a single tool system. The power tool appliance can include a load compensator that is configured to provide force compensation to prevent side-loading on the drive shaft of the power tool. The load compensator can include one or more springs that bias the power tool relative to a housing of the power tool appliance.

(24) The power tool appliance can include a strut that is configured to pivot to align an arm that connects the power tool with the housing of the power tool appliance. The strut can be formed as a plate among other options. The strut is configured to position the arm to provide force compensation and prevent loading on the drive shaft of the power tool.

(25) The arm can connect a mount that connects to the power tool to the housing of the power tool appliance. The arm can be extendable. The arm can be extendable between various lengths such that the power tool appliance can accommodate and operate with power tools of various different sizes and configuration. According to some examples of the disclosure, the arm can be tensioned such that the arm is biased towards a contracted, shortened state.

(26) A drive of the power tool appliance is configured to receive the rotational output from the power tool. An input shaft can extend out of the housing of the power tool appliance to interface with the power tool and receive the rotational output. The input shaft can include a hexagonal exterior contour. The input shaft can, in some examples, be monolithically formed with other components of the drive.

(27) Power tool appliances according to some aspects of the disclosure can be configured for operation in gravity and/or suction feed states. Some examples of power tool appliances can include multiple pump inlets, with a suction inlet configured for suction feed operations and a gravity inlet configured for gravity feed operations. Power tool appliances according to some aspects of the disclosure can be inverted between suction feed and gravity feed states. In the suction feed state, a pump inlet of the pump is oriented vertically downward and the pump is configured to draw spray fluid up from a reservoir against the force of gravity. In the gravity feed state, the pump inlet of the pump is oriented vertically upward such that the fluid flows with gravity to feed the pump.

(28) The power tool is statically connected to the power tool appliance. In some examples, a fixation can secure the power tool to the power tool appliance. The fixation can be formed as a strap that attaches a base of the power tool to the power tool appliance to form the static interface.

(29) Power tool appliances according to the disclosure can, in some examples, be configured to increase a rotational speed of the rotational input received from the power tool. The power tool appliance can be configured such that the rotational motion is used to turn a turbine to generate airflow, such as for high volume, low pressure spraying of paints, stains, finishes, and/or other coatings. The power tool appliance can include overdrive gearing that outputs rotation at a higher speed than received to drive the turbine and generate desired airflow.

(30) Components can be considered to radially overlap when those components are disposed at common axial locations along an axis. A radial line extending orthogonally from axis will extend through each of the radially overlapping components. Components can be considered to axially

overlap when those components are disposed at common radial and circumferential locations relative to the axis. An axial line parallel to the axis will extend through the axially overlapping components. Components can be considered to circumferentially overlap when those components are disposed at common radial distance and axial locations along the axis, such that a circle centered on the axis passes through each of the circumferentially overlapping components.

(31) FIG. 1 is a block diagram of tool system 1. Tool system 1 includes power tool appliance 2 and power tool 14. Output assembly 3, appliance output 4, support assembly 5, drive 6, and biaser 7 of power tool appliance 2 are shown. Support assembly 5 includes arm 8 and mount 9. Power tool 14 includes tool handle 30, tool motor 31, tool output 33, power system 34, and trigger 36.

(32) Power tool appliance 2 is configured to connect to power tool 14 to be operated by power tool 14. Power tool 14 is configured to provide operating power to power tool appliance 2. Power tool 14 can be any type of power tool, particularly a type that outputs rotational motion. A common type of such power tool 14 is an impact driver suitable for driving screws into wood and similar materials. The power tool can also be a drill commonly used with bits to drill into material such as wood, as well as drive screws into such materials. The power tool 14 includes a tool handle 30. The entirety of the tool system 1 can be supported by single hand grabbing handle 30 to support the entirety of the tool system 1 and to operate the tool system 1, while a finger of the hand actuates the trigger 36 of the power tool 14.

(33) As shown, the power tool 14 includes a trigger 36 which upon actuation causes power system 34, such as a battery though it is understood that power tool 14 can include a power cord, of the power tool 14 to operate tool motor 31 within the power tool 14 to output rotational motion from the tool output 33 of the power tool 14. The tool output 33 can be formed as a clamp, such as a bit holder, such as a chuck, collet, sleeve, and/or driverback bit holder. The tool output 33 is configured to provide the rotational output from the power tool 14 to the power tool appliance 2. The tool output 33 can be configured as a hex bit socket. Such a hex bit socket can receive and retain a hex bit, such as by radial clamping and/or detent (e.g., ball) in a groove of the hex bit. The power tool appliance 2 does not include a trigger, only the power tool 14 has a trigger which when pulled causes the power tool appliance 2 to operate (e.g., spray). In the example shown, the power tool appliance 2 does not include any electrical components that facilitate output by the power tool appliance 2.

(34) Output assembly 3 is configured to receive rotational output from the power tool 14 and is configured to provide the output of the tool system 1. The output assembly 3 includes appliance output 4 that is configured to provide the working output from output assembly 3. The appliance output 4 can, in some examples, be a mechanical output that provides work. For example, the appliance output 4 can be configured as a saw, spray output, other fluid output (e.g., gas for blowing or liquid such as for liquid transfer), piston, or other tool element.

(35) The power tool 14 is dynamically, mechanically connected to the output assembly 3 to provide the rotational input to the output assembly 3. The support assembly 5 statically connects the power tool 14 to the output assembly 3. The support assembly 5 and output assembly 3 are connected together. The support assembly 5 and output assembly 3 can be movable relative to each other.

(36) The support assembly 5 can be repositioned relative to the output assembly 3. In some examples, the support assembly 5 is movable relative to the output assembly 3 to adjust an angle between the arm 8 and the housing 11. In some example, the support assembly 5 is movable relative to the output assembly 3 by adjusting a length of the arm 8. The support assembly 5 being movable relative to the output assembly 3 facilitates positioning of power tool 14 relative to power tool appliance 2 and formation of static and dynamic interfaces between the power tool 14 and power tool appliance 2.

(37) Housing 11 can enclose and/or support various other components of power tool appliance 2. Housing 11 can be a polymer housing, among other options. The housing 11 can be a clamshell having two lateral sides, among other options.

(38) Drive 6 is at least partially disposed within housing 11. Drive 6 is configured to receive the rotational input from power tool 14 and provide an output to appliance output 4. The drive 6 is configured to mechanically power appliance output 4. For example, the drive 6 can be connected to a pump to power pumping by the pump. In examples in which the appliance output 4 is a spray output, the pump can be configured to pump spray fluid to a nozzle through which the spray fluid is output at a spray. For example, the nozzle can be configured to atomize the spray fluid into the fluid spray.

(39) Drive 6 receives the rotational output from power tool 14. The mechanical output from drive 6 operates the appliance output 4. In some examples, the drive 6 can be configured to provide a linear output that operates the appliance output 4. For example, the drive 6 can be configured to provide a linear reciprocating output (e.g., to a piston or a saw blade) to cause operation of power tool appliance 2. In some examples, the drive 6 can be configured to provide a rotational output that operates the appliance output 4. For example, the drive can be configured to drive rotation of a turbine, gear set, chain, belt, or other device to operate the appliance output 4.

(40) A rotational input shaft 52 extends from outside of the housing 11 into the housing 11. Input shaft 52 is configured to interface with the power tool 14 to receive the rotational output from the power tool 14. In the example shown, the input shaft 52 interfaces with tool output 33, such as by being received within the tool output 33, to receive the rotational output from the power tool 14. The input shaft 52 is connected to the drive 6 to provide the rotational motion to the drive 6. In some examples, the input shaft 52 can be formed monolithically with other rotating portions of the drive 6. The input shaft 52 can be considered to form a portion of the drive 6. The drive 6 converts the rotational motion to another type of motion, such as a rotational output or reciprocating motion.

(41) Input shaft 52 extends out from a rear side of the housing 11. Input shaft 52 can be multifaceted. The multifaceted exterior of the input shaft 52 facilitates formation of a driving, dynamic interface between power tool appliance 12 and power tool 14. The input shaft 52 can be a hex shaft, having the same hexagonal shape as a hex bit, that is received within the tool output 33 of the power tool 14 (e.g., within the hex bit socket).

(42) It is noted that in the example shown the rotational input shaft 52 is part of the power tool appliance 2 and is not part of the power tool 14. As such, upon disconnection of the power tool appliance 2 from the power tool 14, the rotational input shaft 52 remains with the power tool appliance 2 and not the power tool 14, as the input shaft 52 is withdrawn from the clamp 32. However, in various other examples, the rotational input shaft 52 is not part of the power tool appliance 2.

(43) Support assembly 5 includes arm 8 that is connected to housing 11. Arm 8 extends from housing 11. The arm 8 is movable relative to the housing 11. The arm 8 can be repositioned to change a location of the mount 9 relative to the housing 11. In some examples, the arm 8 is repositionable to change an angle between the arm 8 and the housing 11. In some examples, the arm 8 is repositionable to change a length of the arm 8. Mount 9 is connected to arm 8. Mount 9 is connected to an opposite end of arm 8 from the housing 11. Mount 9 is configured to interface with the power tool 14. The mount 9 can be a plate, amongst other options. The mount 9 can include a first side for interfacing with the base of the power tool 14 (e.g., a battery of the power tool 14 or other portion of a base of the power tool 14) and a second side opposite of the first side that is facing downward away from the power tool 14 and can support tool system 1 on a surface when set down. In some examples, mount 9 is movable relative to arm 8. Mount 9 can be moved relative to arm 8 to reorient mount 9 relative to arm 8.

(44) It is noted that not all examples of power tool appliance 2 includes mount 9 in the manner of the power tool 14 resting on top of the mount 9. In various other examples, the power tool 14 can be held by the mount 9 without the mount 9 forming a tray. For example, the mount 9 may clamp onto the sides of the base of the power tool 14, or may clamp onto the tool handle 30 of the power tool 14, among other options.

(45) The power tool appliance **2** includes one or more biasers **7** that are configured to urge the housing **11** relative to the power tool **14**. In some examples, the biaser **7** is configured to align support assembly **5** and housing **11** relative to each other. The biaser **7** can be configured to set an angle of the support assembly **5** relative to the housing **11**. For example, the biaser **7** can be configured to set an angle θ of the arm **8** relative to vertical. The biaser can be configured to set a length of the support assembly **5**. For example, the biaser can be configured to set a length of arm **8**. Such a biaser **7** can be configured to urge the arm **8** towards a shorter, contracted state. The one or more biasers **7** can be formed as one or more springs, among other options.

(46) In some examples, the one or more biasers **7** of the power tool appliance **2** are configured to preload the dynamic interface between the power tool appliance **2** and the power tool **14** (e.g., the interface between input shaft **52** and tool output **33**). The biaser can preload the dynamic interface by biasing the housing **11** vertically upward. Such a configuration can counteract the weight of output assembly **3** and can align the input shaft **52** with tool output **33**. In some additional or alternative examples, the biaser **7** can preload the dynamic interface by biasing the arm **8** towards a contracted state. For example, the biaser **7** can pull the mount **9** to bias the power tool **14** vertically upward while the dynamic interface is fixed between the input shaft **52** and the power tool **14**. The biaser **7** can preload the dynamic interface by biasing the housing **11** upward and into alignment with the power tool **14**. The biaser **7** can preload the dynamic interface by biasing the support assembly **5** and housing **11** away from each other. For example, the biaser **7** can urge the housing **11** vertically upward and longitudinally away from the support assembly **5** while the dynamic interface is fixed between the input shaft **52** and the power tool **14**. Preloading the dynamic interface provides for a secure connection between power tool **14** and power tool appliance **2** that provides for efficient and easy operation. Such preloading can further inhibit the power tool **14** from entering an impact mode. Power tool appliance **2** provides significant advantages. Biased **7** is configured to urge the output assembly **3** relative to the power tool **14**, providing for a tight fit and alignment therebetween. The biased **7** can include one or more springs. Forming the one or more biasers **7** from one or more springs facilitates automatic and quick fitting and alignment. The springs can retract the arm **8** to a desired length that fits the power tool **14** without requiring additional input from the user, such as adjusting knobs or clamps. The springs can bias the output assembly **3** relative to the support assembly **5** to align the power tool **14** and output assembly **3**. Such a configuration can automatically compensate and maintain alignment even during changes in orientation or weight of the power tool appliance **2**. For example, the weight of power tool appliance **2** can change in examples in which power tool appliance **2** is configured as a sprayer and as spray fluid is emitted. The orientation of the center of gravity of output assembly **3** can change as the appliance output **4** is oriented upward or downward.

(47) FIG. 2A is a first side elevational view of tool system **10**. FIG. 2B is a second side elevational view of tool system **10**. FIGS. 2A and 2B are discussed together. Tool system **10** is substantially similar to tool system **1** (FIG. 1). It is noted that, while tool system **10** is shown as a sprayer, any other type of power tool appliance can be substituted for the sprayer, including use of the features shown on the sprayer, such as for reciprocating a tool element such as a saw.

(48) Tool system **10** includes power tool appliance **12**, power tool **14**, and reservoir **16a**. Output assembly **17**, tip assembly **20**, support assembly **22**, and fixation **24** of power tool appliance **12** are shown. Housing **18** of output assembly **17** is shown. Support assembly **22** includes arm **26** and mount **28**. Power tool **14** includes tool handle **30**, clamp **32**, power system **34**, and trigger **36**. Power tool appliance **12** can be similar to or the same as power tool appliance **2**. Output assembly **17** can be similar to or the same as output assembly **3**. Support assembly **22** can be similar to or the same as support assembly **5**.

(49) Power tool appliance **12** is configured to connect to power tool **14** to be operated by power tool **14**. Power tool **14** is configured to provide operating power to power tool appliance **12**. Power tool **14** can be any type of power tool, particularly a type that outputs rotational motion. A common

type of such power tool **14** is an impact driver suitable for driving screws into wood and similar materials. The power tool can also be a drill commonly used with bits to drill into material such as wood, as well as drive screws into such materials. The power tool **14** includes a tool handle **30**. The entirety of the tool system **10** can be supported by single hand grabbing handle **30** to support the entirety of the tool system **10** and to operate the tool system **10**, while a finger of the hand actuates the trigger **36** of the power tool **14**.

(50) As shown, the power tool **14** includes a trigger **36** which upon actuation causes power system **34**, such as a battery though it is understood that power tool **14** can include a power cord, of the power tool **14** to operate an electric motor within the power tool **14** to output rotational motion from the clamp **32** of the power tool **14**. The clamp **32** of the power tool **14** can be any bit holder option, such as a chuck, collet, sleeve, and/or driverback bit holder. The clamp **32** can be a hex bit socket. Such a hex bit socket can receive and retain a hex bit, such as by radial clamping and/or detent (e.g., ball) in a groove of the hex bit. The power tool appliance **12** does not include a trigger, only the power tool **14** has a trigger which when pulled causes the power tool appliance **12** to operate (e.g., spray).

(51) The power system **34** can be a battery as shown. The power system **34** can be located at a base of the power tool **14** such that the power tool **14** ordinarily rests upon the power system **34**. In particular, the power system **34** can be a detachable, rechargeable battery pack that is mounted to the bottom of the gun handle **30**.

(52) Output assembly **17** is configured to receive rotational output from the power tool **14** and is configured to provide the output of the tool system **10**. In the example shown, the output assembly **17** is configured for fluid spraying, though it is understood that not all examples are so limited. The power tool **14** is dynamically, mechanically connected to the output assembly **17** to provide the rotational input to the output assembly **17**. The support assembly **22**, which can also be referred to as a support, statically connects the power tool **14** to the output assembly **17**. The support assembly **22** and output assembly **17** are connected together. In the example shown the support assembly **22** and output assembly **17** are pivotably connected to each other.

(53) Housing **18** can enclose and/or support various other components of power tool appliance **12**. Housing **18** can be similar to or the same as housing **11**. Housing **18** can be a polymer housing, among other options. The housing **18** can be a clamshell having two lateral sides, among other options. As shown, a rotational input shaft **52** extends from outside of the housing **18** into the housing **18**. As discussed in more detail below, the housing **18** contains a mechanism which converts the rotational motion to another type of motion, such as reciprocating motion for operating a piston, saw, or other tool element.

(54) Input shaft **52** extends out from a rear side of the housing **18**. The input shaft **52** can be a hex shaft, having the same hexagonal shape as a hex bit, that is received within the clamp **32** of the power tool **14** (e.g., within the hex bit socket). Such hexagonal shape includes six flat sides that come together at six corners. The corners can have 120-degree interior angles. The input shaft **52** can include a circumferential groove that allows for interfacing with a detent of the clamp **32** of the power tool **14**. The input shaft **52** can be retained in the clamp **32** in the same manner as a hex bit is retained by the clamp **32** when the hex bit is used for driving fasteners such as screws. For example, the clamp **32** can clamp onto the input shaft **52**.

(55) It is noted that in the example shown the rotational input shaft **52** is part of the power tool appliance **12** and is not part of the power tool **14**. As such, upon disconnection of the power tool appliance **12** from the power tool **14**, the rotational input shaft **52** remains with the power tool appliance **12** and not the power tool **14**, as the input shaft **52** is withdrawn from the clamp **32**. However, in various other examples, the rotational input shaft **52** is not part of the power tool appliance **12**.

(56) Being that the particular example of the power tool appliance **12** shown is part of a sprayer, the power tool appliance **12** includes a spray outlet **38**. The spray outlet **38** can be a nozzle. In the

example shown, spray outlet **38** is formed as a portion of spray tip assembly **20**. Spray tip assembly **20** is connected to housing **18** and supported by housing **18**. The spray outlet **38** is configured to atomize the fluid output by a pump of the power tool appliance **12** into a fluid spray. The spray outlet **38** can be a rotatable spray tip which can reverse its flow for unclogging. Specifically, the spray outlet **38** can be part of a rotatable spray tip **40** that includes a barrel that is received in a cylindrical cavity of a tip housing **42** of the tip assembly **20**, the barrel rotatable within the cylindrical cavity by 180-degrees to reverse the direction of flow through the barrel, to push any clogs out of the spray tip **40**.

(57) Reservoir **16a** is supported by housing **18** such that reservoir **16a** can be carried with housing **18**. The reservoir **16a** is configured to contain a supply of the spray fluid, such as paint. The spray fluid inside of the fluid reservoir **16a** can be drawn up into the housing **18**. As further explained herein, the fluid reservoir **16a** can be located above the housing **18** such that gravity feeds the spray fluid from the fluid reservoir **16a** into the housing **18** or can be located below the housing such that the fluid is sucked from the fluid reservoir **16a** into the housing **18**. The fluid reservoir **16a** can include a squeezable liner inside of a windowed housing, the squeezable liner flexible such that the liner collapses as fluid is withdrawn from the fluid reservoir **16a**. Alternatively, the fluid reservoir **16a** may be rigid cup or pot without a liner that collapses.

(58) Support assembly **22** is configured to support the power tool **14** relative to the housing **18**. The support assembly **22** can thereby support the power tool **14** relative to the input shaft **52**. The support assembly **22** can support the power tool **14** relative to a pump, a drive, and other internals of the housing **18**.

(59) Support assembly **22** includes arm **26** that is connected to housing **18**. Arm **26** can be similar to or the same as arm **8**. Arm **26** extends from housing **18**. Arm **26** is connected to housing **18** at a rear end of housing **18**. The arm **26** connects to the housing **18** at an opposite longitudinal end of housing **18** from spray outlet **38**.

(60) Mount **28** is connected to arm **26**. Mount **28** can be similar to or the same as mount **9**. Mount **28** is connected to an opposite end of arm **26** from the housing **18**. The arm **26** can be considered to connect to housing **18** at an inner end of arm **26** and can be considered to connect to mount **28** at an outer end of arm **26**. Mount **28** is configured to interface with the power tool **14**. The mount **28** can be a plate, amongst other options. The mount **28** can include a first side for interfacing with the base of the power tool **14** (e.g., a battery of the power tool **14** or other portion of a base of the power tool **14**) and a second side opposite of the first side for interfacing with the base of the power tool **14** when the power tool appliance **12** is flipped, as further discussed herein, or that is otherwise facing downward away from the power tool **14**. It is noted that in various examples the mount **28** is intended to only have its first side interface with the base of the power tool **14** such that the second side of the mount **28** is not intended to interface with the base of the power tool **14**, and housing **18** does not flip in the manner discussed herein.

(61) Fixation **24** is configured to secure the power tool **14** to the power tool appliance **12**. The fixation **24** statically connects the power tool **14** to the power tool appliance **12**. In the example shown, the fixation **24** is formed as a strap, though it is understood that other configurations are possible. The strap can include hook-and-loop fasteners. In some examples, the strap can be a double-sided hook-and-loop strap that attaches to itself. For example, both sides of the fixation **24** can include interspersed hooks and loops such that either side can connect to itself or to the other side of fixation **24**. The fixation **24** can attach the power tool **14** to the mount **28**. The fixation **24** can be another type of strap that does not include hook-and-loop connection in various other examples.

(62) In some examples, the fixation **24** is configured to wrap around a portion of the power tool **14** to retain the power tool on the mount **28**. In the example shown, fixation **24** is configured to wrap around the power system **34** of the power tool **14** to retain the power tool **14** on the mount **28**. The fixation **24** wraps around and engages the top and lateral sides of the power system **34** of the power

tool **14** to retain the power tool **14** on the mount **28**. The fixation **24** does not engage the bottom side of the power tool **14** in the example shown, though it is understood that other mounting options are possible. For example, fixation **24** could be a clamp instead of a strap, the clamp clamping onto the power system **34**. In other examples the fixation **24** can interface with the handle **30** or body of the power tool **14**.

(63) In the example shown, fixation **24** secures the power tool **14** to mount **28**. Mount **28** includes slot **44** that extends through mount **28**. Slot **44** is open laterally through mount **28**. The slot **44** extends laterally through the mount **28** with openings on opposite lateral sides of the mount **28** through which the fixation **24** can extend. Slot **44** is configured such that fixation **24** can pass fully through mount **28** within slot **44**. In the example shown, a plurality of slots **44** are formed through mount **28**. The multiple slots **44** facilitate passing fixation **24** around mount **28** at different locations longitudinally along mount **28**. Alternatively, the fixation **24** can wrap all the way around the mount **28**, such that the fixation **24** contacts the second (downward-facing) side of the mount **28**. However, slots **44** accommodating the fixation **24** allows the tool system **10** to be set down on a surface with the second surface of the mount **28** facing down to engage the surface on which the tool system **10** rests while not resting upon the fixation **24**, providing for better balance and preventing tipping or rocking.

(64) In the example shown, the slot **44** is disposed between ribs **46** of the mount **28**. The ribs **46** are disposed on the longitudinal ends of the slot **44** to longitudinally define the slot **44**. The ribs **46** overlap with the fixation **24** to prevent the fixation **24** from sliding longitudinally off of mount **28**. The ribs **46** can also provide additional rigidity to mount **28**, reducing the weight of mount **28**, providing for a lighter and easier to handle power tool appliance **12**.

(65) It is noted that not all examples of power tool appliance **12** includes mount **28** in the manner of the power tool **14** resting on top of the mount **28**. In various other examples, the power tool **14** can be held by the mount **28** without the mount **28** forming a tray. For example, the mount **28** may clamp onto the sides of the base of the power tool **14**, or may clamp onto the tool handle **30** of the power tool **14**, among other options.

(66) In the example shown, arm **26** is movable relative to housing **18**. Arm **26** can be moved relative to housing **18** to reorient arm **26** relative to housing **18**. In the example shown, arm **26** is pivotably connected to housing **18**. Arm **26** is connected to housing **18** by arm pivot **48**. Arm pivot **48** can also be referred to as an upper pivot or an inner pivot. Pivoting can be facilitated by pins, axles, and/or bearings which allow relative rotation. While arm pivot **48** is connected to housing **18**, it is understood that in various examples the arm pivot **48** indirectly connects the arm **26** to the housing **18**. In the example shown, the arm pivot **48** directly connects the arm **26** to the housing **18**. Arm pivot **48** allows the arm **26** to move by rotation relative to the housing **18**.

(67) In the example shown, mount **28** is movable relative to arm **26**. Mount **28** can be moved relative to arm **26** to reorient mount **28** relative to arm **26**. In the example shown, mount **28** is pivotably connected to arm **26**. The mount **28** can rotate relative to arm **26** to move relative to arm **26**. Mount **28** is connected to arm **26** by mount pivot **50**. Mount pivot **50** can also be referred to as a lower pivot or an outer pivot. Pivoting can be facilitated by pins, axles, and/or bearings which allow relative rotation. While mount **28** and arm **26** are rotatably supported in the example shown, it is understood that some examples do not include the arm pivot **48** and/or the mount pivot **50**.

(68) In some examples, the arm pivot **48** can be configured as a free pivot in which arm **26** can freely swing from housing **18** when a power tool **14** is not connected. In other examples, the arm pivot **48** can include resistance that maintains the arm **26** in a desired orientation until overcome by a reorienting force, such as the user grasping and reorienting arm **26**. Similar to arm pivot **48**, the mount pivot **50** can be configured as a free pivot in which mount **28** can freely swing from arm **26** when a power tool **14** is not connected. In other examples, the mount pivot **50** can include resistance that maintains the mount **28** in a desired orientation until overcome by a reorienting force, such as the user grasping and reorienting mount **28**.

(69) FIG. 3 is an isometric view of power tool appliance 12. FIG. 4A is an isometric cross-sectional view of power tool appliance 12 taken along line 4-4 FIG. 3. FIG. 4B is an elevational cross-sectional view of power tool appliance 12 taken along line 4-4 in FIG. 3. FIG. 5 is a side elevational view of tool system 10. FIGS. 2A, 2B, 3, 4A, 4B and 5 are discussed together. Housing 18, tip assembly 20, support assembly 22, input shaft 52, drive 56, pump 58, spray valve 62, connector 64, shaft aligner 66, and reservoir 16a are shown. Front side 68, rear side 70, reservoir side 72, and lateral side 72 of housing 18 are shown. Spray outlet 38, spray tip 40, and tip housing 42 of tip assembly 20 are shown. Arm 26 and mount 28 of support assembly 22 are shown. Upper arm extension 76, lower arm extension 78, and tensioner 126 of arm 26 are shown. Drive shaft 80, drive body 82, outer ring 84, balls 86, and output 88 of drive 56 are shown. Pump body 90, piston 92, pump inlet 94, cylinder 96, and outlet valve 98 of pump 58 are shown. Lid 100, cup 102, and liner of reservoir 16a are shown. Longitudinal directions X-X, lateral directions Y-Y, and vertical directions Z-Z are indicated in FIG. 3.

(70) Housing 18 supports and/or encloses and/or connects together other components of power tool appliance 12. Drive 56 is at least partially disposed within housing 18. Drive 56 is configured to convert a rotational input from the power tool 14 into a linear reciprocating output. The reciprocating output from the drive 56 is provided to the pump 58 to power pumping by the pump 58. Drive 56 can connect directly to the power tool 14, such as via input shaft 52, in some examples. Drive 56 can be configured similar to or the same as drive 6.

(71) Drive shaft 80 is configured to receive the rotational input from the power tool 14. In the example shown, the drive shaft 80 forms part of the drive 56 and thus participates in converting rotational motion to linear reciprocating motion. In the example shown, drive 56 is formed as a wobble drive, though it is understood that various other types of drives are possible, such as eccentric and crank arm.

(72) The drive shaft 80 can form at least part of a bearing which converts continuous rotational motion to irregular motion due to the tilted orientation and positioning of the outer ring 84 on the drive shaft 80. Balls 86 are disposed between drive shaft 80 and outer ring 84. The balls 86 ride in corresponding grooves on the drive shaft 80 and the outer ring 84. The grooves are canted relative to a plane orthogonal to the drive axis DA. Such a configuration causes the outer ring 84 to rock as the drive shaft 80 rotates on the drive axis DA. Such rocking causes the output 88 of the drive to reciprocate and thus causes reciprocation of the piston 92 of the pump 58. In the example shown, the output 88 is formed as a projection that extends from the outer ring 84, though it is understood that not all examples are so limited.

(73) Drive shaft 80 is rotatably supported by bushings 128a, 128b. Bushings 128a, 128b are disposed on opposite axial sides of the bearing interface between drive shaft 80 and outer ring 84. The bushings 128a, 128b are disposed on opposite axial sides of the drive body 82 of the drive shaft 80 along the axis DA of the drive 56. The drive body 82 forms an enlarged portion of the drive shaft 80 configured to interface with the balls 86 and including the groove that the balls 86 ride in. The input shaft 52 extends in axial direction AD2 from the drive body 82. The input shaft 52 is cantilevered from the rearward bushing 128b. The rearward bushing 128b is disposed axially between the input shaft 52 outside of housing 18 and the drive body 82.

(74) Input shaft 52 is connected to drive 56. The input shaft 52 can be considered to form a portion of the drive 56. Input shaft 52 extends out a rear side 70 of housing 18 such that input shaft 52 can extend into a power tool 14 to connect to the power tool 14. Input shaft 52 is configured to form a dynamic mechanical connection with the power tool 14 to receive rotational input from the power tool 14. In the example shown, the drive shaft 80 also forms the rotational input shaft 52 which extends outside of the housing 18. The input shaft 52 is cantilevered out from the rear side 70 of the housing 18.

(75) The drive shaft 80 and input shaft 52 are formed monolithically in the example shown. As such, in various examples, a single piece of metal (e.g., steel) that forms the input shaft 52 also

forms part of the drive **56** and participates in converting rotational motion into reciprocating motion, instead of merely transmitting rotational motion from one area to another area. Having the input **51** (including the drive shaft **80** and input shaft **52**) of the drive **56** be one piece simplifies manufacturing and allows this part to be made from higher strength material being that this part must both interface with a variety of different power tools **14**, side load and/or preload, transmit rotational torque, and directly participate in converting rotational motion into linear reciprocating motion.

(76) The drive **56** converts rotational motion received from the power tool **14** into linear reciprocating motion. The linear reciprocating motion can reciprocate the piston **92** to pump the fluid, such as providing suction to move the fluid from the reservoir **16a** and further driving the fluid through spray valve **62** and out the spray outlet **38**. The fluid exits from within housing **18** through the front side **68** of housing **18**. While drive **56** is shown as being operatively connected to a piston **92**, it is understood that drive **56** can be connected to any desired fluid displacer configured to reciprocate to pump the fluid, such as a piston or diaphragm.

(77) It is noted that the drive **56** does not include any gearing or gears in the example shown. Further, the entirety of the power tool appliance **12** does not include any gears. Therefore, each rotation of the input shaft **52** by the power tool **14** results in one reciprocation of the piston **92**, in a 1:1 ratio. A single rotation of the input shaft **52** can cause the piston **92** to displace in both a forward axial direction **AD1** and through a pressure stroke and a rearward axial direction **AD2** and through a suction stroke, thereby causing the piston **92** to move through one pump cycle. It is understood, however, that various other embodiments may include gearing.

(78) Pump **58** is supported by housing **18**. Pump **58** can be at least partially disposed within housing **18**. In some examples, pump **58** is fully disposed within housing **18**. Pump body **90** supports other components of pump **58**. Pump body **90** further defines fluid pathways for fluid to flow between reservoir **16a** and spray outlet **38**. In some examples, the fluid output from reservoir **16a** is received directly into the pathways within pump body **90**, the fluid is then output from the pump body **90** and to the valve assembly **60** that includes spray valve **62** before flowing downstream to and through spray outlet **38**.

(79) Piston **92** is at least partially disposed within pump body **90**. Piston **92** is configured to reciprocate on pump axis **PA** to pump the spray fluid from reservoir **16a** and downstream through spray outlet **38**. Piston **92** is connected to drive **56** to receive the reciprocating output from drive **56**. Piston **92** is connected to output of drive **56** to receive the reciprocating motion from drive **56**. Piston **92** is configured to displace in rearward axial direction **AD2** during a suction stroke in which fluid is drawn from reservoir **16a** and into pump **58**. Piston **92** is configured to displace in forward axial direction **AD1** during a pressure stroke to pressurize the fluid and drive the fluid downstream under pressure for emission through spray outlet **38**. While pump **58** is shown as a piston pump, it is understood that not all examples are so limited. Further, while pump **58** is shown as a single piston pump, it is understood that not all examples are so limited. For example, pump **58** can be configured to include a plurality of pistons **92** (e.g., two, three, or more). The multiple pistons can be reciprocated out of phase with respect to each other to pump the spray fluid. Each of the multiple pistons can be connected to a separate output **88** of the drive **56** to be reciprocatingly displaced by the drive **56**.

(80) The fluid contacting portion of piston **92** is connected indirectly to the drive **56** in the example shown. A forward portion of piston **92**, which can be metallic, reciprocates and contacts the fluid to pump the fluid. A rearward portion of piston **92**, which can be formed from polymer, interfaces with output **88** to receive the reciprocating motion from output **88**. The irregular motion of the output **88** forward and rearward (while the outer ring **84** is not allowed to rotate with the drive shaft **80**) pushes and pulls the piston **92** forward and rearward in a reciprocating motion to pump the fluid, or provide another reciprocating function in alternative examples.

(81) Cylinder **96** is disposed within pump body **90**. The piston **92** reciprocates within cylinder **96** to

pump the fluid. The cylinder **96** can, in some examples, be formed separately from the pump body **90**. For example, the cylinder **96** can be formed from metal, such as tungsten carbide, while pump body **90** can be formed from polymer. Cylinder **96** can be considered to define a pump chamber in which the spray fluid is pressurized during the pumping stroke of the piston **92**.

(82) Outlet valve **98** is disposed downstream of the piston **92**. Outlet valve **98** is disposed downstream of the pump chamber. Outlet valve **98** is configured as a one-way valve that allows spray fluid to flow downstream from the pump chamber while preventing retrograde flow to the pump chamber. In the example shown, the outlet valve **98** is configured as a spring-biased normally-closed ball check valve.

(83) Spray valve **62** is disposed downstream of piston **92**. Spray valve **62** is disposed downstream of pump **58**. Spray valve **62** is normally closed. Valve housing **106** is mounted to pump body **90**. Spray valve **62** is disposed within valve housing **106**. In the example shown, the spray valve **62** includes a valve spring **114** that maintains the spray valve in a closed state. The spray valve **62** can be configured such that the fluid pressure reaching a threshold overcomes the force of the spring and causes the spray valve **62** to open, allowing spray fluid to flow downstream to and through spray outlet **38**.

(84) Tip assembly **20** is mounted to valve housing **106** in the example shown. Tip housing **42** is mounted to valve housing **106** by collar **54** in the example shown. Spray tip **40** is rotatably mounted within tip housing **42**. Tip housing **42** is rotatable relative to collar **54** on spray axis SA, which is the axis along which the spray outlet **38** is configured to output the spray fluid. Spray outlet **38** is formed as a nozzle through the spray tip **40**.

(85) Pump **58** is configured to receive spray fluid through pump inlet **94**. The piston **92** is reciprocated by the drive on the piston axis PA. The piston axis PA is radially offset from the drive axis DA in the example shown. Reciprocation of the piston **92** draws spray fluid from reservoir **16a** through pump inlet **94**. The spray fluid is pressurized and driven downstream through outlet valve **98**, spray valve **62**, and spray outlet **38** to be sprayed on the target surface. In the example shown, the piston **92**, spray valve **62**, and spray outlet **38** are disposed coaxially on the piston axis PA such that the spray axis SA and piston axis PA are disposed coaxially, though it is understood that not all configurations are so limited. In some examples, pump **58** includes at least one piston **92** disposed coaxially with the spray outlet **38** and includes one or more other pistons disposed to reciprocate on axes offset from the spray axis SA.

(86) The piston **92** is vertically offset from the drive axis DA. In some examples, the piston **92** can be offset from the drive **56** and be disposed to radially overlap with the rotating components of the drive **56**.

(87) In some examples, the piston **92** and drive **56** can be vertically aligned. For example, the drive axis DA and the axis PA can be disposed coplanar in an X-Z plane. Such a configuration can provide for efficient force transmission while also providing for a compact configuration of power tool appliance **12**.

(88) The piston **92** is disposed vertically below the drive **56** with the power tool appliance **12** in the suction feed state. The piston **92** is disposed vertically below the drive axis DA with the power tool appliance **12** in the suction feed state.

(89) The piston **92** is disposed vertically above the drive **56** with the power tool appliance **12** in the gravity feed state (FIGS. 7A-7B). The piston **92** is disposed vertically above the drive axis DA with the power tool appliance **12** in the suction feed state.

(90) Reservoir **16a** is configured to store a supply of the spray fluid for use during spraying. Reservoir **16a** is supported by housing **18**. In the example shown, the reservoir **16a** is indirectly connected to housing **18**. Reservoir **16a** is directly connected to pump body **90** in the example shown to connect to and be supported by housing **18**.

(91) Cup **102** is configured to house the spray fluid. In some examples, a collapsible liner **104** can be disposed in cup **102** and the fluid can be within the collapsible liner **104**. The collapsible liner

104 can collapse towards pump inlet **94** during operation. In other examples, the reservoir **16a** does not include a collapsible liner **104**. In such examples, a suction tube can extend from pump inlet **94** to or proximate the bottom end of the reservoir **16a**. Lid **100** is connected to cup **102**. Lid **100** can connect the collapsible liner to cup **102**. Lid **100** can connect reservoir **16a** to power tool appliance **12**.

(92) The power tool appliance **12** includes a connector **64**. The connector **64** can connect directly to the reservoir **16a**. The connector **64** can include threading, bayonet structure, tabs or slots (for complement tab-and-slot mating) or other fixation complementary with that of the reservoir **16a**. In the example shown, the connector **64** is formed as a portion of the pump body **90**. The connector **64** interfaces with complementary structure formed in the lid **100** of the reservoir **16a** in the example shown. Pump inlet **94** extends from connector **64** in the examples shown, though it is understood that not all examples are so limited. In various other examples, the pump inlet **94** and connector **64** can be separated. The connector **64** can attach to reservoir **16a** or other type of fluid reservoir. The connector **64** in this example is a bayonet type, however threading, press fit, lever, tab and groove, or other options are possible.

(93) Support assembly **22** is connected to housing **18**. Support assembly **22** is configured to statically interface with the power tool **14** to statically connect the power tool **14** to the power tool appliance **12**. The support assembly **22** can connect the power tool appliance **12** and power tool **14** together such that the power tool **14** can provide the rotational input to the power tool appliance **12** at the dynamic interface with input shaft **52**. In the example shown, the support assembly **22** is movable relative to housing **18** and is reconfigurable to accommodate different sizes and configurations of power tool **14**.

(94) Arm **26** is connected to housing **18** at arm pivot **48**. In the example shown, the arm **26** extends along an arm axis AA that is repositionable relative to both the housing **18** and mount **28**. Arm **26** extends between and connects housing **18** and mount **28**. Mount **28** is connected to an outer end of arm **26** at mount pivot **50**. Mount **28** can be repositioned relative to arm **26** by pivoting at mount pivot **50**.

(95) Upper arm extension **76** is connected to housing **18**. Upper arm extension **76** includes arm yoke **108** that extends on both lateral sides **72** of the housing **18** to connect to the housing **18** at the arm pivot **48**. Lower arm extension **78** is connected to mount **28**. Lower arm extension **78** includes mount yoke **110** that extends on both lateral sides of the mount **28** to connect to the mount **28** at the mount pivot **50**. The upper arm extension **76** and lower arm extension **78** are movable relative to each other to adjust the length of the arm **26**.

(96) In the example shown, arm **26** is repositionable between a contracted state and an extended state. In some examples, the arm **26** can be considered to be infinitely positionable between the contracted state and the extended state, as arm **26** can be adjusted to any position between and including positions associated with the contracted and extended states. In the example shown, the upper arm extension **76** and the lower arm extension **78** are disposed in a telescoping configuration, though it is understood that not all examples are so limited. Upper arm extension **76** extends within lower arm extension **78** and the upper arm extension **76** and lower arm extension **78** are configured to relatively slide to change the length of the arm **26**. Upper arm extension **76** is received within the lower arm extension **78** and the upper arm extension **76** moves within the lower arm extension **78**, although this receiving convention can be reversed. Such movement allows the arm **26** to be lengthened or shortened to accommodate power tools **14** of different sizes.

(97) Tensioner **126** is configured to urge arm **26** towards the contracted or shortened state. Tensioner **126** is formed as a spring in the example shown. Tensioner **126** is disposed within arm **26** in this example, though it is understood that not all examples are so limited. Tensioner **126** is connected to both upper arm extension **76** and lower arm extension **78** to urge the lower arm extension **78** towards housing **18**. The tensioner **126** is shown as a spring, such as a metal spring, but it is understood that tensioner **126** can be formed in any suitable configuration for urging arm

26 towards the contracted state, such as an elastic band among other options.

(98) Tensioner **126** can maintain the arm **26** at the desired length to accommodate whichever power tool **14** is connected to power tool appliance **12**. For example, the lower arm extension **78** can be pulled to a length that accommodates the power tool **14** and the power tool **14** can be connected to the mount **28**, such as by the fixation **24** and before or after pulling the arm **26** to the desired length. The tensioner **126** pulls the lower arm extension **78** towards the housing **18**. The mount **28** being connected to the power tool **14** limits such retraction of the lower arm extension **78** while the tensioner **126** continues to pull the lower arm extension **78** towards the housing **18**, thereby loading the static interface for a tight fit between power tool **14** and power tool appliance **12**. Such a configuration can prevent sag and play between power tool **14** and power tool appliance **12**, providing for more ergonomic and efficient operation.

(99) Shaft aligner **66** is configured to bias the output of power tool **14** that connects to the input shaft **52** (e.g., clamp **32**) into coaxial alignment with the input shaft **52** on drive axis DA. Such a configuration maintains concentricity and inhibits side-loading of the output of the power tool **14**, which can prevent the power tool **14** from going into an impact driving mode (e.g., when power tool **14** is formed by an impact driver).

(100) In the example shown, shaft aligner **66** includes strut **112** that is supported by housing **18**. Strut **112** is movable relative to housing **18**. In the example shown, strut **112** is mounted to housing at strut pivot **116** such that strut **112** can pivot on and relative to housing **18**. Strut **112** is configured to interface with arm **26** to bias arm **26** vertically upwards and rearwards relative to input shaft **52**. Strut **112** is configured to interface with arm **26** to bias arm **26** vertically upwards and rearwards relative to housing **18**.

(101) Springs **114** interface with strut **112** to bias strut relative to housing **18**. Springs **114** form biasers (e.g., biasers **7**) that are configured to bias the support assembly **22** and housing **18** relative to each other. The power tool appliance **12** includes at least one biaser that is configured to bias the housing **18** and support assembly **22** to align the input shaft **52** with the power tool **14**.

(102) Springs **114** are disposed between strut **112** and housing **18** in the example shown. The springs **114** are configured to resist movement of the arm **26** towards the housing **18**. In the example shown, shaft aligner **66** includes upper spring **114a** and lower spring **114b** disposed on opposite vertical sides of the drive axis DA. It is understood that, in various examples, the shaft aligner **66** can include a plurality of the upper springs **114a** and/or a plurality of the lower springs **114b**. While shaft aligner **66** is shown as including both upper springs **114a** and lower springs **114b**, it is understood that not all examples are so limited. For example, shaft aligner **66** may include only a single upper or lower spring **114**, such as in configurations in which power tool appliance **12** is not configured to invert between suction and gravity feed configurations.

(103) Strut **112** is configured to interface with arm **26**. Springs **114** resist movement of strut **112**, and thus of arm **26** and the power tool **14** mounted to arm **26**, vertically downward and forward relative to drive axis DA. Strut **112** can be considered to resist movement of the housing **18** vertically downward and rearward relative to drive axis DA to maintain housing **18** and thus drive **56** and input shaft **52**, aligned with the output of power tool **14**. Such a configuration can inhibit side-loading that can be caused due to the weight of the spray fluid contained in the spray reservoir **16a**.

(104) As shown, the arm yoke **108** wraps around both lateral sides of the arm mount **118** and a pin rotationally connects the arm **26** to the arm mount **118** to facilitate rotation of the arm **26** relative to the housing **18**. The strut **112** is mounted on a strut mount **120**. The strut mount **120** can be a narrower portion of the housing **18**, or can be a separate structure. The strut mount **120** can be defined in part by a ridge **122** against which the strut **112** can bottom out against when pivoting, however other options are possible. In the example shown, ridge **122** is oriented transverse to a horizontal X-Y plane. Ridge **122** is non-orthogonal to the horizontal X-Y plane. Ridge **122** is sloped such that ridge **122** is oriented to face vertically upward to resist movement of the strut **112**.

In the example shown, a plurality of ridges **122** are formed due to power tool appliance **12** being invertible. The multiple ridges **122** are disposed in a chevron configuration on the lateral side **74** of the housing **18**. The chevron tip is oriented rearward. The ridges **122** extend vertically and longitudinally forward relative to the tip of the chevron.

(105) In the example shown, strut **112** is a ring-like structure which defines a strut aperture **124**. In such a structure, the rotational input shaft **52** can extend through the aperture **124**. Furthermore, a part of the housing **18** can extend through the aperture **124**. Arm mount **118** extends within aperture **124** in the example shown. The arm mount **118** extends through aperture **124** and is configured to interface with the arm pivot **48**. The arm mount **118** can be part of the housing **18** or can be separate from the housing **18**. The arm mount **118** can include an attachment to the arm **26**, such as one or more pins which facilitate fastening while allowing hinge motion.

(106) The views of FIGS. **2A**, **2B**, **3**, **4A**, **4B** and **5** demonstrate several aspects which can enhance the function of the tool system **10** or other type of power tool appliance **12**. Such features can be useful for getting a proper fit of the power tool **14** with the power tool appliance **12** considering that a single power tool appliance **12** should be usable with a large variety of slightly different types of power tools **14**. In particular, such different types of power tools **14** can be different dimensions and thus aligned with the rotational input shaft **52** slightly differently. Moreover, impact drivers and various other types of power tools **14** operate best when loaded with an axial force along the rotational input shaft **52**. For example, impact drivers can (mechanically) enter an impact function, which can include unwanted hammering action, depending on load transmitted through the rotational input shaft **52**, which is also dependent on the orientation of the power tool **14** relative to the rotational input shaft **52**. Such hammering action of the impact function may be ideal for drilling screws into wood as the intended application of the power tool **14**, but may not be ideal for spraying applications or other uses of the power tool appliance **12**, and is ideally avoided. Side-loading of the rotational input shaft **52** can encourage entry into impact function. Various features of power tool appliance **12** can help facilitate proper loading, and avoid such side-loading or other factors which contribute to entry into impact function, such as by keeping the hammer of the power tool **14** from slipping, to avoid or minimize entry into the impact function, as further discussed herein.

(107) As shown in FIG. **5**, the weight of the power tool appliance **12** exerts a downward force F_d that is experienced at the dynamic mechanical interface between power tool appliance **12** and power tool **14**. The downward force F_d can vary during operation, such as due to the volume of fluid within reservoir **16a** decreasing during spray operations. The downward force F_d is generated by the weight of the output assembly **17**. The center of gravity F_{cd} exerts downward force, which can vary. The location of the center of gravity F_{cd} can shift longitudinally due to the changing fluid volume in the example shown. In the example shown, the center of gravity F_{cd} is disposed on an opposite longitudinal side of the shaft aligner **66** from the support assembly **22**.

(108) An upward force F_y is generated that counteracts the downward force F_d exerted by the power tool appliance **12**. The upward force F_y concentrically aligns the rotational output from the power tool **14** and the rotational receiver (e.g., input shaft **52**) of the output assembly **17**. A pre-tensioner is configured to exert upward forces F_y that counteract the downward force F_d , preventing side-loading. In the example shown, shaft aligner **66** includes springs **114** such that the upward force F_y is variable to account for the variable downward force F_d .

(109) The arm **26** can adjust in length, to be longer or shorter. Tensioner **126** urges the arm **26** to be shorter. Such tensioning can exert force F_b that has an upward component to maintain loading between the power tool appliance **12** and the power tool **14** which avoids entry into the impact function. The tensioner **126** is attached to each of the upper arm extension **76** and the lower arm extension **78** to urge them together and/or toward greater overlap. It is noted that tensioner **126** can also maintain pressure on the power tool **14** via the rotational input shaft **52** to maintain proper load. This is because the tensioner **126** pulls the mount **28** (generally) toward the input shaft **52**,

which due to the base of the power tool **14** being fixed to the mount **28**, this urges the power tool **14** vertically upward, which helps to preload the power tool **14** and counteract the downward force F_d of the power tool appliance **12**. Such pressure when properly maintained can help avoid entry into the impact function.

(110) While the arm **26** includes tensioner **126**, various other features are provided which can help avoid entry into impact function in a complementary or independent fashion with respect to the tensioner **126**. It is noted that the tensioner **126** may not be included in various embodiments.

(111) As previously mentioned, the arm **26** pivots with respect to the housing **18**. Such pivoting can provide different alignments of the power tool **14** with respect to the rotational input shaft **52** and the housing **18**. The pivoting can change the angle θ between arm **26** and a vertical axis VA. Such variation in the angle θ can affect the force exerted by shaft aligner **66**, such as by strut **112**.

Furthermore, pressure can be maintained on the arm **26** to maintain, or be urged into, an orientation which fosters proper alignment of the power tool **14** relative to the rotational input shaft **52**. Shaft aligner **66** exerts force F_a that is upward and forward to bias the housing upward and forward relative to the power tool **14**. Such force F_a can counteract at least a portion of the downward force F_d exerted by the power tool appliance **12**. During operation, the user holds the power tool **14** by the handle **30** of the power tool **14**. The power tool **14** and power tool appliance **12** are held stationary relative to each other, such that the power tool **14** does not cause rotation of the power tool assembly **12**, by the static interface between housing **18** and power tool **12** formed through support assembly **22**. The power tool **14** is also dynamically connected to the output assembly **17** at the dynamic interface with the input shaft **52**. The force F_d pulls down on the output assembly **17** to pull housing **18**, and thus the input shaft **52** that is supported within housing **18**, in rotational direction RD1 about arm pivot **48**. The shaft aligner **66** exerts a counteracting force F_a that resists pivoting of housing **18** and urges output assembly **17** in rotational direction RD2. The vertical force components can cancel out to prevent side loading on the driveshaft of the power tool **14**.

(112) The power tool appliance **12** includes one or more biasers (e.g., springs **114**, tensioner **126**) that are configured to align support assembly **22** and housing **18** relative to each other. The biaser can be configured to set an angle of the support assembly **22** relative to the housing **18**. For example, the biaser can be configured to set an angle θ of the arm **26**. In the example shown, the biaser that sets angle of the arm **26** is formed by spring **114**. The biaser can be configured to set a length of the support assembly **22**. For example, the biaser can be configured to set a length of arm **26**. In the example shown, the biaser that sets the length of the arm **25** is formed by tensioner **126**. The biaser can thereby align the power tool **14** and housing **18** to align the input shaft **52** with the power tool **14** output.

(113) The one or more biasers of the power tool appliance **12** are configured to preload the dynamic interface between the power tool appliance **12** and the power tool **14**. The biaser can preload the dynamic interface by biasing the power tool **14** vertically upward. The biaser can preload the dynamic interface by biasing the arm **26** towards a contracted state. For example, the tensioner **126** can pull the mount **28** to bias the power tool **14** vertically upward while the dynamic interface is fixed between the input shaft **52** and the power tool **14**. The biaser can preload the dynamic interface by biasing the housing **18** upward and into alignment with the power tool **14**. The biaser can preload the dynamic interface by biasing the support assembly **22** and housing **18** away from each other. For example, the springs **114** can bias the housing **18** vertically upward and longitudinally away from the support assembly **22** while the dynamic interface is fixed between the input shaft **52** and the power tool **14**. Preloading the dynamic interface provides for a secure connection between power tool **14** and power tool appliance **12** that provides for efficient and easy spray operations.

(114) In the example shown, a pivot urges alignment. The pivot can be considered to urge the arm **26** or the housing **18** relative to the other one of the arm **26** and the housing **18**. The pivot can be strut **112**. Strut **112** can be considered to form the pivot plate. Strut **112** can be a support. The pivot

can be urged, such as by springs **114**. In this particular embodiment, the strut **112** engages the arm **26** as urged by the spring **114**. The spring **114** urges the strut **112** away from, or rotationally upward, with respect to the housing **18**. This engages the arm **26** and urges the arm **26** rotationally upwards, which supports and aligns the power tool **14** connected to the arm **26** (e.g., via the mount **28**). While shaft aligner **66** is shown as including strut **112**, it is understood that not all examples are so limited. For example, shaft aligner **66** can include one or more discrete projections or other structure that is spring biased to exert the force F_a on output assembly **17**.

(115) Spring **114** pushes off from housing **18** or other structure to urge the strut **112** away from the housing **18**. The strut **112** is rotationally fixed directly to the housing **18** or indirectly relative to the housing **18** (e.g., in either case via pivoting connecting such as via pins, axles, bearings, and/or other type of connection which facilitates relative rotation while still retaining the strut **112**) such that the urging by the spring **114** pushes the strut **112** to rotate about the strut pivot **116**. It is noted that the strut **112** may not pivot and may instead translate linearly or other motion, such that pivoting of the strut **112** is not the only range of motion that various different struts may undergo. The strut **112** urges, directly by direct contact or indirectly in the case of an intermediary component, the housing **18** as the strut **112** itself is urged by the spring **114**. The arm **26** is pivotally connected directly to the housing **18** or pivotally fixed with respect to the housing **18** (and indirectly connected to the housing **18**) such that the force F_a urges the arm **26** and/or housing **18** to swing or otherwise rotate upwards toward the input shaft **52** by the force F_a . Being that the power tool **14** is supported by the arm **26**, or indirectly connected to the arm **26**, such urging of the housing **18** pushes the output assembly **17** upwards. Such pushing can optimally position the power tool **14** and/or preload the power tool **14** via the input shaft **52**.

(116) In the example shown, the upper springs **114a** and lower springs **114b** can work in concert to generate the biasing force F_a . With power tool appliance **12** in the suction feed orientation (shown in FIG. 5), the lower springs **114b** can push the strut **112** in rotational direction RD1 about the pivot of the strut **112**. The lower springs **114b** can be considered to push the housing **18** in rotational direction RD2. The upper springs **114a** can be connected to strut **112** and housing **18** such that upper springs **114a** pull on housing **18** to also exert force to urge housing **18** in rotational direction RD2. With power tool appliance **12** in the gravity feed orientation (shown in FIGS. 7A and 7B), the upper springs **114a** can push the strut **112** in rotational direction RD2 about the strut pivot **116**. The lower springs **114b** can be considered to push the housing **18** in rotational direction RD1. The lower springs **114b** can be connected to strut **112** and housing **18** such that lower springs **114b** pull on housing **18** to also exert force to urge housing **18** in rotational direction RD1. The upper springs **114a** and lower springs **114b** can thus work together to generate the biasing force F_a .

(117) The combination of the upward force components of the forces F_a and F_b can combine to form the upward force F_y that balances with the downward force F_d . Such a configuration inhibits side-loading on the input shaft **52** and power tool **14**, maintaining power tool **14** in a desired operating mode and decreasing the force required by power tool **14** to drive input shaft **52**, which can allow for longer battery life and thus longer continuous operation of power tool appliance **12**.

(118) When the power tool **14** is mounted, the tensioner **126** and the springs **114** can work together to find a balance and proper alignment between the power tool **14** and the rotational input shaft **52**. In some examples, the spring forces can oppose each other. As such, the tensioner **126** and spring **114** can be in partial opposition to each other as the tensioner **126** pulls the power tool **14** nearer to the housing **18** and the spring **114** urges the power tool **14** upwards and partially away from the housing **18** due to the swinging motion of the arm **26**. It is understood, however, that not all examples are so limited.

(119) The power tool appliance **12** is connected to mount **28** to form a static connection between power tool **14** and power tool appliance **12**. For example, fixation **24** can extend through slot **44** in mount **28** and be wrapped around the base of power tool **14** to fix power tool **14** on mount **28**, among other connection options. The power tool **14** is connected to power tool appliance **12** by a

dynamic connection interface that mechanically powers operation of power tool appliance **12**. The input shaft **52** interfaces with the power tool **14** at the dynamic mechanical interface to receive the rotational output from the power tool. As discussed above, the shaft aligner **66** and tensioner **126** can bias power tool **14**, via arm **26**, to align input shaft **52** and power tool **14** during operation. (120) During operation, the power tool **14** is powered to provide a continuous rotational output in a single rotational direction. It is understood that, in the example shown, the power tool **14** can be caused to rotate clockwise or counterclockwise and will cause pumping by pump **58** and spraying of fluid in either configuration.

(121) The rotation from the power tool **14** is provided to the input shaft **52** which is rotated on the drive axis DA. The input shaft **52** and drive body **82** rotate together, such as due to being formed monolithically. The rotating drive body **82** causes reciprocation of the drive output **88**, due to the canted bearing races between drive body **82** and outer ring **84** in this example. In the example shown, the drive **56** extends along the drive axis DA. The input shaft **52** is configured to receive the rotational input for drive **56**. The drive body **82** is configured to rotate to provide an output from drive **56**, rotation of drive body **82** causing reciprocation of the piston **92**. In the example shown, the rotational input and the rotational output of the drive **56** are disposed coaxially. In the example shown, the rotational input and the rotational output of the drive **56** are contiguous.

(122) Power tool appliance **12** provides significant advantages. In the example shown, the power tool appliance **12** provides a sprayer that can be operated by multiple different power tools **14** having different configurations. Power tool appliance **12** being operable by a power tool **14**, such as a drill or impact driver, allows for the power tool **14** to perform multiple functions (e.g., powering the power tool appliance **12** for the function of the power tool appliance **12** and drilling or screwing in or out), while also decreasing the components and thus cost of the fluid-handling power tool appliance **12** relative to a sprayer having an integrated motor. In the example shown, the power tool appliance **12** does not include any electronics. The electronic components of tool system **10** are disposed on and remain with power tool **14**, at a location isolated from the fluid pathways through power tool appliance **12**, protecting such components from fluid contamination.

(123) Power tool appliance **12** contains drive **56** that converts the rotational output from power tool **14** into linear reciprocating motion that powers pumping by the pump **58**. Input shaft **52** includes a contoured end, hexagonal in the example shown, that facilitates formation of a keyed interface with the power tool **14** for transmitting rotational motion to the input shaft **52**. The input shaft **52** is thereby contoured to receive rotational output from a variety of different power tools **14**. The input shaft **52** and drive body **82** are unitary in the example shown, which increases the strength of the drive shaft **80** and can provide for greater operational life for power tool appliance **12**.

(124) Support assembly **22** facilitates mounting of various different power tools **14** having different sizes and configurations to power tool appliance **12**. Arm **26** is extendable to allow the output of the power tool **14** to align with the input shaft **52** while the power tool **14** can rest on and be connected to mount **28**. Tensioner **126** urges arm **26** towards the contracted state while also allowing arm **26** to be extended to any position intermediate and including the contracted state and the extended state. In some examples, support assembly **22** can include length fixers, such as clamps, fasteners, detents, etc., that fix the length of arm **26** even if the static connection between power tool appliance **12** and power tool **14** is broken.

(125) Arm **26** can be reoriented relative to housing **18**, further facilitating accommodation of multiple power tools **14** having different configurations. Mount **28** can be reoriented relative to arm **26** which also assists in accommodating multiple different power tools **14** having multiple different configurations.

(126) The mount **28** includes multiple slots **44** through which a fixation **24** can be passed to wrap around the base of the power tool **14** and statically connect the power tool **14** to the power tool appliance **12**. The static connection facilitates conversion of the rotational output to linear driving motion for driving pump **58** by preventing the power tool appliance **12** from rotating relative to

power tool **14**.

(127) Shaft aligner **66**, and in some examples tensioner **126**, align power tool appliance **12** and power tool **14** to prevent side-loading at the dynamic interface between power tool appliance **12** and power tool **14**. The shaft aligner **66** exerts a biasing force on arm **26** that counteracts the weight of power tool appliance **12** to align input shaft **52** and the output of power tool **14** coaxially. The tensioner **126** can further exert biasing force for alignment. Such a configuration prevents side-loading and can thereby prevent the power tool **14** from entering into an undesirable impact mode.

(128) The strut **112** in the illustrated examples is able to engage the arm **26** from either a standard or inverted orientation, the standard orientation associated with the suction feed state and the inverted orientation associated with the gravity feed state. In the way, the strut **112** can engage the arm **26** from the either side of the arm **26** and can always rotate upwards to swing the arm **26** upwards regardless of whether the housing **18** is orientated for suction or gravity feed.

(129) FIG. **6** is an isometric view of power tool appliance **12** with power tool **14** removed and configured in a suction feed state. FIG. **7A** is an isometric view of power tool appliance **12** with power tool **14** removed and configured in a gravity feed state. FIG. **7B** is an isometric view of power tool appliance **12** in the gravity feed state similar to FIG. **7A** but with reservoir **16b** mounted. FIGS. **6**, **7A** and **7B** are discussed together.

(130) Power tool appliance **12** is operable in a suction feed state, in which fluid flows upward into pump **58**, and a gravity feed state, in which fluid flows downward into pump **58**. In the example shown, the power tool appliance **12** is invertible between the suction feed state (FIG. **6**) and the gravity feed state (FIGS. **7A** and **7B**). In the suction feed state, the pump inlet **94** is oriented vertically downward. In the gravity feed state, the pump inlet **94** is oriented vertically upward.

(131) The reservoir-facing side **72** of housing **18**, from which the pump inlet **94** extends, is oriented vertically downward with the power tool appliance **12** in the suction feed state. The reservoir **16a** extends outward and vertically downward from the pump inlet **94** with power tool appliance **12** in the suction feed state. The reservoir **16a** depends from the connector **64** with the power tool appliance **12** in the suction feed state. In the example shown, the reservoir **16a** hangs from and is supported by pump body **90**, which forms connector **64**, with power tool appliance **12** in the suction feed state.

(132) The reservoir-facing side **72** of housing **18**, from which the pump inlet **94** extends, is oriented vertically upward with the power tool appliance **12** in the gravity feed state. The reservoir **16b** extends outward and vertically upward from the pump inlet **94** with power tool appliance **12** in the gravity feed state. The reservoir **16b** is supported on the connector **64** with the power tool appliance **12** in the gravity feed state. In the example shown, the reservoir **16a** rests on and is supported by pump body **90**, which forms connector **64**, with power tool appliance **12** in the gravity feed state.

(133) While power tool appliance **12** is shown as including a single pump inlet **94** that is repositionable between being oriented vertically upward and vertically downward, it is understood that not all examples are so limited. For example, power tool appliance **12** can include multiple pump inlets **94** such that the user can select whether to attach a reservoir **16a** to one inlet for suction feed or reservoir **16b** to another inlet for gravity feed. The power tool appliance **12** can include a selector for selectively connecting the multiple inlets to the spray valve **62** and then spray outlet **38**.

(134) The arm **26** and housing **18** are repositionable relative to each other such that pump inlet **94** can be oriented vertically upwards or downwards. The repositioning facilitates reconfiguration of the power tool appliance **12** between the suction feed and gravity feed states. In the example shown, the arm **26** and/or housing **18** can be pivoted relative to the other one of the arm **26** and/or housing **18** such that housing **18** is oriented with the pump inlet **94** vertically upward or vertically downward.

(135) In the example shown, arm **26** includes arm side **130a** that is oriented longitudinally towards the power tool **14** with the power tool appliance **12** in the suction feed state and includes arm side

130b that is oriented longitudinally away from power tool **14** with the power tool appliance **12** in the suction feed state. The arm side **130a** is oriented longitudinally away from the power tool **14** with the power tool appliance **12** in the gravity feed state and the arm side **130b** is oriented longitudinally towards from power tool **14** with the power tool appliance **12** in the gravity feed state.

(136) The arm **26** is disposed longitudinally between the pump inlet **94** and the trigger **36** of the power tool **14**. The arm **26** is disposed directly longitudinally between reservoir **16a** and handle **30** with the power tool appliance **12** in the suction feed state. The arm **26** is disposed longitudinally between the reservoir **16b** and the handle **30** with the power tool appliance **12** in the gravity feed state. The arm **26** can directly longitudinally overlap with the handle **30** and/or trigger **36** with the power tool appliance **12** in the gravity feed state. The support assembly **22** can thereby provide a trigger guard that can prevent inadvertent triggering of the power tool **14** by extending around an area within which the trigger **36** is disposed with power tool appliance **12** in either of the suction or gravity feed states.

(137) Mount **28** includes interface sides **132a**, **132b**. Interface sides **132a**, **132b** are configured to interface with power tool **14** to support the power tool **14** on power tool appliance **12**. In some examples, the interface side **132a**, **132b** not interfacing with power tool **14** can rest on a support surface, such as the floor or a table, to support the tool system **10** on that support surface. The interface sides **132a**, **132b** of the mount **28** are formed as opposite sides of the mount **28**. The slots **44** are disposed vertically between the interface sides **132a**, **132b** with the mount **28** disposed to support the power tool **14**. The ribs **46** extend between interface sides **132a**, **132b** to support interface sides **132a**, **132b**.

(138) With power tool appliance **12** in the suction feed configuration, the interface side **132a** is oriented face up. The interface side **132a** is oriented upwards such that the base of the power tool **14** can rest on interface side **132a**. With the power tool appliance **12** in the gravity feed configuration, the interface side **132b** is oriented face up. The interface side **132b** is oriented upwards such that the base of the power tool **14** can rest on interface side **132b**. The inverting or flipping of the power tool appliance **12** changes which surface of the mount **28** supports the power tool **14**. In this way, the housing **18** is flipped such that the pump inlet **94** faces upward instead of downward. Likewise, the mount **28** is flipped such that a different surface supports the power tool **14** and the other surface can engage a resting surface when the system is set down.

(139) The inversion of the power tool appliance **12** changes the orientation of the pump inlet **94**. With the power tool appliance **12** in the suction feed state, the pump inlet **94** is orientated to face downward for a suction mode in which the fluid is sucked up from the reservoir **16a**. In the gravity feed state, the pump inlet **94** faces upwards to receive the fluid via gravity feed, although the pump **58** within the housing **18** can still provide some amount of suction to move the paint through the pump inlet **94**. The flipping of the orientation of the inlet **94** can be advantageous because different types of fluid reservoirs and fluids operate better or worse in gravity or suction feed, and the flipping of the housing **18** allows the orientation of the pump inlet **94** to be changed based on the best type of feed for the application.

(140) As shown in FIG. 7B, a different type of reservoir **16b** can be connected to the power tool appliance **12** at the connector **64**. Reservoir **16b** is configured for gravity feed. The body of reservoir **16b** narrows towards pump inlet **94** to funnel the spray fluid towards pump inlet **94**. As such, a different type of fluid reservoir **16b** can be attached to feed fluid to the power tool appliance **12** via the pump inlet **94**.

(141) In the example shown, the connector **64** includes a plurality of tabs **136** that are configured to interface with corresponding slots on a reservoir **16a**, **16b** to connect the reservoir **16a**, **16b** to the power tool appliance **12**. The tabs **136** can be evenly arrayed about the pump body **90**. Multiple different configurations of reservoir can be mounted to power tool appliance **12** at connector **64**. As such, power tool appliance **12** allows the user to connect the variation of reservoir that works best

for the spray operations being conducted.

(142) As such, different suction or gravity feed options are possible based on inversion of the power tool appliance **12**. Rotation of the housing **18** and the mount **28** relative to the arm **26** via the arm pivot **48** and mount pivot **50** facilitate such inversion in the example shown. In the example shown, the arm **26** is configured to pivot relative to the housing **18** on a first pivot axis and the mount **28** is configured to pivot relative to the arm **26** on a second pivot axis. In the example shown, the first and second pivot axes are parallel to each other, which facilitates quick and easy reconfiguration between the suction and gravity feed states.

(143) During flipping of power tool appliance **12**, the rear side **70** moves within the arm yoke **108** and between the branches **134** of the arm yoke **108**. The arm mount **118**, and in some examples housing **18**, move within arm yoke **108** and directly between branches **134**. The input shaft **52** that extends out of the rear side **70** passes through the gap between the branches **134** of the arm yoke **108**. In the example shown, the power tool appliance **12** is configured such that the pump inlet **94** does not pass through the arm yoke **108** or between the branches **134**. The power tool appliance **12** is configured such that the reservoir-facing side **72** and front side **68** do not pass through the arm yoke **108** or between the branches **134**. Arm yoke **108** can be considered to form a U-joint in the example shown. The housing **18** is connected to the arm **26** such that input shaft **52** that extends rearward relative to rear side **70** of the housing **18** moves through the arm yoke **108** during reconfiguration between the suction feed state and the gravity feed state.

(144) During flipping, the mount **28** moves within the mount yoke **110**. While mount yoke **110** is shown as formed by arm **26** in the example shown, it is understood that not all examples are so limited. For example, mount yoke **110** can be formed by mount **28** and can extend about or within a portion of arm **26**. Mount yoke **110** can be considered to form a U-joint in the example shown.

(145) During flipping, the housing **18** and mount **28** can be rotated in the same rotational direction relative to arm **26** to reconfigure the power tool appliance **12**. The housing **18** and mount **28** can be rotated in rotational direction RD2 to reconfigure power tool appliance **12** from the suction feed state to the gravity feed state. The housing **18** and mount **28** can be rotated in rotational direction RD1 to reconfigure power tool appliance **12** from the gravity feed state to the suction feed state.

(146) Power tool appliance **12** provides significant advantages. Power tool appliance **12** can be reconfigured between various states based on the operating requirements of the user. In the example shown, the power tool appliance **12** can be reconfigured between the suction and gravity feed states as desired for spray operations. The pump inlet **94** can be reoriented between receiving fluid that is flowing downward and receiving fluid that is flowing upward. In various other examples, the power tool appliance **12** can include other structure or features that are disposed at or near the location of pump inlet **94**. For example, a handle can project from the reservoir-facing side **72** of housing **18** and the user may desired to grip the power tool appliance **12** on the lower or upper side of power tool appliance **12**.

(147) FIG. **8** is an isometric view of drive **56**. FIG. **9** is a cross-sectional view taken along line 9-9 in FIG. **8**. FIGS. **8** and **9** are discussed together. Driveshaft **80**, input shaft **52**, drive body **82**, balls **86**, outer ring **84**, output **88**, bushing **128a**, and bushing **128b** are shown.

(148) Drive **56** is configured to receive a rotational input via input shaft **52** and to output a linear reciprocating motion via output **88**. The drive **56** can receive a continuous rotational input in either rotational direction on the drive axis DA and will provide the reciprocating motion via output **88**. While drive **56** is shown as including a single output **88**, it is understood that drive **56** can include multiple outputs **88** such that drive **56** can cause reciprocation of multiple fluid displacers, such as pistons or diaphragms. The multiple outputs **88** can be arrayed about the drive axis DA. The multiple outputs **88** can be evenly arrayed about the drive axis DA.

(149) Input shaft **52** is configured to interface with the power tool **14** to receive the rotational motion from the power tool **14**. Input shaft **52** includes keyed exterior **138** that is configured to interface with the power tool **14**. The keyed exterior **138** facilitates transmission of the rotational

motion from the power tool **14** to the drive **56**.

(150) In the example shown, the input shaft **52** includes head **140**, shaft groove **142**, and shaft body **144**. The input shaft **52** is elongated along drive axis DA. Shaft body **144** extends axially from driveshaft **80**.

(151) Shaft groove **142** is disposed axially between shaft body **144** and head **140**. Shaft groove **142** can extend between and connect shaft body **144** and head **140**. Shaft groove **142** can receive a detent; such as a ball, shaft, etc.; of the power tool **14** that axially locates the input shaft **52** relative to the power tool **14**. It is understood that not all examples include shaft groove **142**.

(152) Head **140** is disposed at a distal, free end of the input shaft **52**. In the example shown, the keyed exterior **138** is formed on both the shaft body **144** and the head **140**, though it is understood that not all examples are so limited.

(153) Drive body **82** is a radially enlarged portion of driveshaft **80**. Drive body **82** is disposed axially between bushings **128a**, **128b**. Drive body **82** can be captured axially between bushings **128a**, **128b** to prevent driveshaft **80** from displacing axially along drive axis DA.

(154) Outer ring **84** is disposed about drive body **82**. Outer ring **84** is prevented from rotating about the drive axis DA with driveshaft **80**. Instead, rotation of the driveshaft **80** causes the outer ring **84** to wobble such that output **88** reciprocates linearly.

(155) Inner groove **146** is formed on drive body **82**. Inner groove **146** is canted relative to a plane orthogonal to the drive axis DA. Inner groove **146** extends transverse to a plane orthogonal to the drive axis DA. The inner groove **146** can extend fully annularly about the drive axis DA. Outer groove **148** is formed on outer ring **84**. Outer groove **148** can be disposed coaxially with structure of outer ring **84**.

(156) Balls **86** are disposed between drive body **82** and outer ring **84**. Balls **86** are disposed in inner groove **146** and outer groove **148**. Rotation of the drive body **82** can displace the balls **86** axially forward and rearward due to rotation of the inner groove **146**, thereby causing the outer ring **84** to wobble such that outputs **88** move forward and rearward. The drive body **82** can be considered to form an inner race while the outer ring **84** forms an outer race of a bearing that provides the linear input to the fluid displacer of the pump **58**.

(157) Bushings **128a**, **128b** are configured to rotationally support the drive **56**. The bushings **128a**, **128b** can support the drive **56** on the housing **18**. While bushings **128a**, **128b** are shown, it is understood that drive **56** can be supported in any desired manner on housing **18**. For example, bushings **128a**, **128b** can be formed as roller bearings or other structure configured to support relative rotational motion.

(158) Bushing **128b** is disposed axially between drive body **82** and input shaft **52**. Bushing **128b** is disposed axially between the bearing interface between outer ring **84** and drive body **82** and the input shaft **52**. Bushing **128b** is disposed axially between output **88** and input shaft **52**. The input shaft **52** is cantilevered relative to bushing **128b**. The output **88** is disposed axially between bushings **128a**, **128b**. The output **88** being disposed axially between bushings **128a**, **128b** can balance loads transmitted to output **88** and thus to pump **58** to prevent side-loading of the piston **92** of the pump **58**.

(159) In the example shown, input shaft **52** is formed as a unitary part of driveshaft **80**. Monolithic structure forms both the inner race of the bearing (e.g., the drive body **82** forming inner groove **146**) and the keyed exterior **138** of the input shaft **52**. The drive body **82** and input shaft **52** can be formed from the same piece of metal (e.g., steel) in the example shown. Drive **56** can be configured such that a contiguous metallic piece receives the rotation from the power tool **14** and exerts force that transmits the rotation (e.g., by inner groove **146** interfacing with balls **86**) to the linear reciprocating component of the drive **56**.

(160) The material that forms the input shaft **52** can be contiguous with material that forms at least part of an eccentric of the drive **56**. In the example shown, the drive body **82** can be considered to form the eccentric of the drive **56**. In the example shown, the drive shaft **80** is not symmetric about

its axis DA on which the bearing is mounted (i.e. it is radially thicker in some areas and radially thinner elsewhere). The input shaft **52** in the example shown is contiguous with the radially thicker and thinner portions of the drive shaft **80**. The input shaft **52** is contiguous with the portion of drive shaft **80** forming the inner race of the bearing, in the example shown.

(161) Drive **56** provides significant advantages. The input shaft **52** being monolithically formed with other rotational input components of drive **56** provides for a stronger and more rigid input to drive **56**. Such a configuration provides for increased life and thereby decreases costs. Drive **56** is configured to directly interface with the power tool **14** to receive the input from the power tool **14**, which provides for a relatively simple and easy to use configuration. Keyed exterior **138** facilitates transmission of rotational motion from a standard power tool (e.g., drill or impact driver) to the power tool appliance **12**. The keyed exterior **138** can be received by the power tool **14** and can receive torque while preventing relative rotation to provide the rotational output from the power tool **14** to the drive body **82**.

(162) FIG. **10A** is a first isometric view of shaft aligner **66**. FIG. **10B** is a second isometric view of shaft aligner **66**. FIGS. **10A** and **10B** are discussed together. In the example shown, shaft aligner **66** includes strut **112** and springs **114**. Springs **114** are shown connected to strut **112**. In the example shown, the strut **112** interfaces with upper springs **114a** and lower springs **114b**.

(163) The strut **112** interfaces with a plurality of springs **114** which are configured to push and/or pull on the strut **112**. The strut **112** includes face **150a** and face **150b**. Face **150a** is configured to engage with arm side **130a** of arm **26**, such as with power tool appliance **12** in the suction feed configuration. Face **150b** is configured to engage with arm side **130b** of arm **26**, such as with power tool appliance **12** in the gravity feed configuration. As previously mentioned, the strut **112** may only include one of the faces **150a**, **150b** and may not include the other one of faces **150a**, **150b**, particularly in examples in which the power tool appliance **12** is not reconfigurable to flip between states.

(164) Face **150a** is sloped such that a thickness T of the strut **112** increases between a central portion of the strut **112** and edge **152a** of the strut **112**. Similarly, face **150b** is sloped such that the thickness T of the strut **112** increases between the central portion of the strut **112** and edge **152b** of the strut **112**. The sloped configurations of the faces **150a**, **150b** orient the faces **150a**, **150b** vertically upward when the face **150a**, **150b** interfaces with the arm **26**. Such a configuration provides the upward force compensation that facilitates alignment between the power tool **14** and input shaft **52**.

(165) Springs **114** are disposed on an inner side **154** of strut **112**. Springs **114** can be connected to strut **112** such that springs **114** can exert pulling in addition to pushing forces on the strut **112**. In the example shown, upper springs **114a** interface with strut portion **156a** and lower springs **114b** interface with strut portion **156b**. Face **150a** is formed on an outer side of strut portion **156a**. Face **150b** is formed on an outer side of strut portion **156b**. As discussed above, upper springs **114a** and lower springs **114b** are configured to interface between housing **18** and strut **112** to bias strut **112**, and thus arm **26** and power tool **14**, to provide an upward force compensation that accounts for the weight of power tool appliance **12**.

(166) The shaft aligner **66** including springs **114** provides significant advantages. The springs **114** are configured to balance the power tool appliance **12** relative to the power tool **14** to maintain concentricity therebetween. The springs **114** provide a variable force compensation that accounts for the changing weight of the power tool appliance **12** as spray fluid is emitted during operation. As the weight of the power tool appliance **12** drops, the force required to be exerted by springs **114** also drops, which allows springs **114** to expand while still compensating for the weight of power tool appliance **12**.

(167) Aperture **124** is formed through strut **112** in the example shown. In this embodiment, the strut **112** is a ring-like structure which defines aperture **124**. In such a structure, the rotational input shaft **52** can extend through the aperture **124** in the strut **112**. Furthermore, a part of the housing **18** (e.g.,

arm mount **118**) can extend through the aperture **124**. Aperture **124** facilitates positioning the strut pivot **116** longitudinally between the arm pivot **48** and the center of gravity (which can be aligned on reservoir **16a**, **16b**) of the power tool appliance **12**. Such a configuration facilitates the shaft aligner **66** compensating for the weight of power tool appliance **12** to align power tool appliance **12** with power tool **14**.

(168) In the example shown, rocker **158** is formed on the back side of strut **112**. The strut **112** is configured to rock about rocker **158** to push rearward either the face **150a** or face **150b**, with the plurality of springs **114** applying opposing forces on opposite sides of the rocker **158**, which balances the strut **112** and thereby the arm **26** to facilitate loading of the rotational input shaft **52** to avoid entry into impact function. It is noted while a single strut **112** is shown, two different pivoting structures can instead be employed, respectively defining face **150a** and face **150b**. Also, a single pivoting structure including only a single face **150a**, **150b** may be included, particularly in embodiments in which the housing **18** does not invert for alternative suction and gravity feeds. Strut **112** can engage the arm **26** from the either side of the arm **26** and can always rotate upwards to swing the arm **26** upwards regardless of whether the housing **18** is orientated for suction or gravity feed.

(169) In some examples, strut **112** may not be directly connected to housing **18**. For example, the arm **26** can interface with strut **112** to engage the rocker **158** on projections of the housing **18**. The arm **26** can prevent the strut **112** from moving off of the housing **18**. In some examples, the springs **114** are not connected to housing **18**. In such an example, the shaft aligner **66** can be a single assembly, such as formed by the strut **112** and springs **114**, that is not directly connected to, but is held on, the housing **18**.

(170) FIG. **11A** is an isometric view of support assembly **22** in a contracted state. FIG. **11B** is an isometric view of support assembly **22** in an extended state. FIGS. **11A** and **11B** are discussed together. Arm **26** and mount **28** of support assembly **22** are shown. Upper arm extension **76**, lower arm extension **78**, arm yoke **108**, and mount yoke **110** of arm **26** are shown. Slots **44**, ribs **46**, and interface sides **132a**, **132b** of mount **28** are shown.

(171) Arm **26** is repositionable between various lengths to facilitate connection of different power tools **14** having different sizes and/or configurations to power tool appliance **12**. Upper arm extension **76** and lower arm extension **78** are relatively movable along an arm axis AA to vary the length of arm **26**. In the example shown, upper arm extension **76** and lower arm extension **78** are telescopically arranged, though it is understood that not all configurations are so limited. The upper arm extension **76** and lower arm extension **78** can be keyed together, such as by having non-circular cross-sections orthogonal to arm axis AA, to prevent relative rotation between the arm extensions **76**, **78**. In the example shown, the upper arm extension **76** extends into the lower arm extension **78** such that an amount of the upper arm extension **76** disposed outside of the lower arm extension **78** varies as the length of arm **26** varies. In various other examples, the lower arm extension **78** can extend within the upper arm extension **76**, among other non-telescoping options.

(172) Upper arm extension **76** is configured to mount the arm **26** to housing **18**. In some examples, upper arm extension **76** can connect directly to housing **18**. In the example shown, the upper arm extension **76** is configured to mount to housing **18** by the arm yoke **108**. The branches **134** of the arm yoke **108** are configured to wrap around opposite lateral sides of an arm mount **118**, which can be formed by part of the housing **18**, to connect to the arm mount **118** and thus to the housing **18**. In the example shown, the arm yoke **108** connects to the arm mount **118** by a pivot, such as a pin or other structure that allows relative rotation.

(173) Mount **28** is connected to lower arm extension **78**. In some examples, the mount **28** can be connected directly to lower arm extension **78**. In the example shown, the mount **28** is configured to connect to lower arm extension **78** by the mount yoke **110**. The mount yoke **110** extends from lower arm extension **78**. The mount yoke **110** is formed as a portion of the lower arm extension **78** in the example shown. The mount **28** extends within the mount yoke **110**. The branches **160** of the

mount yoke **110** are configured to wrap around opposite lateral sides of mount **28**. In the example shown, the mount yoke **110** connects to the mount **28** by at mount pivot **50**, such as a pin or other structure that allows relative rotation.

(174) Mount **28** includes interface sides **132a**, **132b**. As discussed above, mount **28** can be flipped such that either interface side **132a**, **132b** is oriented vertically upward to support the power tool **14** while the other interface side **132a**, **132b** can be oriented vertically downward such as to support the power tool appliance **12** on a surface. Slots **44** extend laterally through the mount **28**. Ribs **46** are disposed between the slots **44**. Slots **44** provide one or more locations for the fixation **24** to pass through mount **28** to wrap around mount **28** and the base of the power tool **14** to statically fix the power tool **14** to the power tool appliance **12**. In the example shown, the mount **28** includes ribs **46** at the longitudinal ends of the mount **28** such that each slot **44** is closed vertically in both vertical directions and longitudinally in both longitudinal directions. The ribs **46** can be laterally elongate.

(175) FIG. **12** is an isometric view of power tool appliance **12'**. Power tool appliance **12'** is substantially similar to power tool appliance **12** (best seen in FIGS. **2A**, **2B**, **3**, **4A**, **4B** and **5**), except that power tool appliance **12'** includes support assembly **22'** instead of support assembly **22**. Same components between power tool appliance **12'** and power tool appliance **12** include the same reference numbers and it is understood that components of power tool appliance **12'** and power tool appliance **12** can be the same unless explicitly described as otherwise.

(176) Support assembly **22'** extends in the rearward axial direction AD2 from the rear side **70** of housing **18**. Support assembly **22'** is configured to interface with the power tool **14** to statically connect the power tool **14** to the power tool appliance **12'**. In the example shown, the support assembly **22'** includes support arm **162** and brackets **164a**, **164b**.

(177) Support arm **162** extends longitudinally rearward from housing **18**. Support arm **162** is connected to housing **18** by mount ring **172**, which can extend fully or partially about housing **18** and can, in some examples, clamp onto housing **18**. Support arm **162** is configured to extend over the power tool **14**. Brackets **164a**, **164b** are connected to support arm **162**. Each bracket **164a**, **164b** includes retainer **166**, locators **168**, and brace **170**. The braces **170** hang below the support arm **162**. The support assembly **22'** is configured to receive a body of the power tool **14** between the brackets **164a**, **164b**. Specifically, the body of the power tool **14** is configured to be positioned between braces **170** of the brackets **164a**, **164b**. Locators **168** extend from the brace **170** and into the support arm **162** to locate the bracket **164a**, **164b** on the support arm **162**. The retainers **166** are configured to allow the brackets **164** to shift laterally inwards towards each other to clamp the body of the power tool **14** while preventing the brackets **164** from shifting laterally outward unless the retainer **166** is unlocked. In the example shown, the retainers **166** are formed as ratchet straps. The retainers can be unlocked and the brackets **164** slid laterally outward to break the static interface and allow dismounting of the power tool **14**.

(178) FIG. **13** is an isometric view of power tool appliance **12''**. Power tool appliance **12''** is substantially similar to power tool appliance **12** (best seen in FIGS. **2A**, **2B**, **3**, **4A**, **4B** and **5**) and power tool appliance **12'** (FIG. **12**), except that power tool appliance **12''** includes support assembly **22''** instead of support assembly **22** or support assembly **22'**. Same components between power tool appliance **12''** and power tool appliance **12** include the same reference numbers and it is understood that components of power tool appliance **12''** and power tool appliance **12** can be the same unless explicitly described as otherwise.

(179) Support assembly **22''** extends in the rearward axial direction AD2 from the rear side **70** of housing **18**. Support assembly **22''** is configured to interface with the power tool **14** to statically connect the power tool **14** to the power tool appliance **12''**. In the example shown, the support assembly **22''** includes inner bracket **174** and outer bracket **176**. The inner bracket **174** and outer bracket **176** can be considered to form a mount **28'** of the support assembly **22''**.

(180) Support assembly **22''** extends longitudinally rearward from housing **18**. Support assembly **22''** is configured to receive power tool **14** within the space defined by the inner bracket **174** and

outer bracket **176**. The inner bracket **174** is connected to housing **18**. The outer bracket **176** is connected to and supported by inner bracket **174**. The inner bracket **174** and outer bracket **176** are configured to longitudinally clamp the power tool **14** on the power tool appliance **12''**. The outer bracket **176** can be moved in forward axial direction **AD1** to exert the clamping force. Fasteners **178** can be fixed to maintain the clamping force and secure the power tool **14** to the power tool appliance **12''**. In the example shown, the outer bracket **176** and inner bracket **174** are telescopically arranged, with the outer bracket **176** received within the inner bracket **174**, though it is understood that not all examples are so limited.

(181) FIG. **14** is a diagram illustrating a tool system **10'** including power tool appliance **12'''**. Power tool appliance **12'''** is substantially similar to power tool appliance **12** (best seen in FIGS. **2A**, **2B**, **3**, **4A** and **4B**), power tool appliance **12'** (FIG. **12**), and power tool appliance **12''** (FIG. **13**), except that power tool appliance **12'''** does not include a piston. Power tool appliance **12'''** is configured as a high volume, low pressure (HVLP) sprayer which uses the flow of air from turbine **180** to atomize and spray paint and other coatings. The drive **56'** includes an overdrive gear set **182** which increases the rotational speed from the input via the input shaft **52** to a higher RPM which is delivered to the turbine **180**. In the example shown, the drive **56'** does not convert rotational motion to linear reciprocation motion. Instead, the drive **56'** transmits and increases the speed of the rotational motion output from the power tool **14** to rotatably drive the turbine **180**. For example, the overdrive gear set **182** can increase the rotational speed of the turbine up to 30 k rpm, among other options.

(182) The airflow from the turbine **180** is routed through narrowing duct **184**. Such airflow can be partially routed to reservoir **16'** to force up fluid and/or the speed of the air can suck up fluid from the reservoir **16'**. In either case, the fluid is atomized by the air and exits the spray outlet **38** as a spray fan. It is understood that, in some examples, power tool appliance **12''** can be reconfigured between suction and gravity feed states.

Discussion of Non-Exclusive Examples

(183) The following are non-exclusive descriptions of possible examples according to various aspects of the disclosure.

(184) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance including a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; and a mount connected to the housing, the mount configured to interface with the power tool to statically connect the power tool to the housing.

(185) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(186) A shaft aligner connected to the housing, the shaft aligner configured to exert a biasing force between the housing and the power tool.

(187) The shaft aligner includes at least one spring.

(188) The shaft aligner is configured to bias the housing away from the mount.

(189) A support assembly connected to the housing, the support assembly including the mount; wherein the shaft aligner interfaces with the support assembly.

(190) The support assembly further comprises an arm, the arm extending between and connecting the housing and the mount.

(191) The arm is pivotable relative to the housing.

(192) The mount is pivotable relative to the arm.

(193) The arm includes an arm spring configured to tension the arm and the arm spring configured to bias the arm into a contracted state.

(194) The arm includes an upper arm and a lower arm, the upper arm connected to the housing and

the lower arm connected to the mount.

(195) The lower arm is movable relative to the upper arm to move the arm between the contracted state and an elongated state.

(196) The shaft aligner is configured to exert the biasing force rearward relative to the housing.

(197) The shaft aligner is configured to exert the biasing force vertically upward relative to the housing.

(198) The shaft aligner is configured to exert the biasing force rearward relative to the housing and vertically upward relative to the housing.

(199) The shaft aligner includes: a strut; and a spring configured to bias the strut outward from the housing.

(200) The shaft aligner further comprises a second spring configured to bias the strut outward from the housing.

(201) The spring is configured to urge the strut in opposition to the second spring.

(202) The strut is configured to pivot relative to the housing.

(203) The strut comprises an upper face and a lower face, the strut rocking to alternately extend or withdraw the upper face and the lower face.

(204) A support assembly connected to the housing, the support assembly including the mount; wherein the strut interfaces with the support assembly.

(205) The support assembly further comprises an arm, the arm extending between and connecting the housing and the mount, wherein the strut is configured to interface with the arm to orientate the arm.

(206) A fixation configured to selectively attach a base of the power tool to the mount.

(207) The fixation is formed as a strap.

(208) The strap includes hook and loop fasteners.

(209) The strap extends through a slot in the mount.

(210) The slot is longitudinally bracketed by a first rib and a second ribs.

(211) The slot is disposed vertically between a first side of the mount and a second side of the mount such that the slot is closed vertically towards the power tool and away from the power tool.

(212) The slot is a first slot of a plurality of slots through the mount.

(213) Each slot of the plurality of slots is open laterally through the mount.

(214) The mount is formed as a plate.

(215) The plate comprises a first mounting surface and a second mounting surface located on an opposite side of the plate with respect to the first mounting surface.

(216) The plate is configured to flip to support the power tool on alternatively the first mounting surface and the second mounting surface.

(217) The fixation includes double-sided hook-and-loop fasteners such that the fixation is configured to remain within the slot during flipping of the power tool and is configured to attach the power tool on either of the first mounting surface and the second mounting surface.

(218) An arm extending from the housing, the arm connecting the mount to the housing.

(219) The arm is pivotable relative to the housing.

(220) The mount is pivotable relative to the arm.

(221) The arm is extendable between a contracted state and an extended state to change a length of the arm.

(222) An arm spring configured to bias the arm towards the contracted state.

(223) An input shaft that extends outward from the housing, the input shaft including a hexagonal exterior contour, the input shaft configured to be received within the power tool and to transmit rotational motion from the power tool to the drive.

(224) The input shaft is part of a single piece of metal which also converts, at least in part, rotational motion to reciprocating motion to drive an output of the power tool appliance.

(225) The drive is a wobble drive including a drive body configured to rotate and a wobble plate

including a drive projection, the drive projection configured to reciprocate due to rotation of the drive body; and the input shaft extends from the drive body and is disposed coaxially with an axis of rotation of the drive body.

(226) The input shaft is formed monolithically with the drive body.

(227) The drive includes a plurality of balls disposed between the drive body and the wobble plate.

(228) The drive projection is connected to a pump supported by the housing to cause pumping by the pump.

(229) The housing is configured to be inverted to convert a pump inlet of a pump supported by the housing between a suction state in which the pump is configured to receive fluid via suction when the pump inlet is facing vertically downward and a gravity state in which the pump is configured to receive the fluid via gravity feed when the pump inlet is facing upward.

(230) The pump inlet includes a connector, the connector configured to interface with a reservoir to connect the reservoir to the housing.

(231) The connector is connectable to a first reservoir of a first configuration and to a second reservoir of a second configuration different from the first configuration.

(232) An arm extending between and connecting the housing and the mount, wherein the arm is pivotable on the housing such that the pump inlet can pivot between the suction state and the gravity state.

(233) The mount is pivotable relative to the arm such that a first mounting surface of the mount is oriented vertically upward with the pump inlet in the suction state and such that a second mounting surface of the mount is oriented vertically upward with the pump inlet in the gravity state.

(234) A first face of the arm is oriented longitudinally away from the mount with the pump inlet in the suction state and the first face is oriented longitudinally towards the mount with the pump inlet in the gravity state.

(235) The arm includes an upper clevis connected to the housing.

(236) An input shaft extending out from the housing through a rear side of the housing is configured to pass through the upper clevis during repositioning between the suction state and the gravity state.

(237) A portion of the housing is disposed within the upper clevis.

(238) The arm includes a lower clevis, the mount connected to the arm within the lower clevis.

(239) An arm extending between and connecting the housing and the mount, wherein the arm is extendable between a contracted state in which the arm has a first length and an extended state in which the arm has a second length greater than the first length.

(240) The arm is positionable to have a plurality of different lengths between the first length and the second length.

(241) The arm includes an arm tensioner configured to urge the arm towards the contracted state.

(242) The arm tensioner is disposed within the arm.

(243) The arm tensioner comprises a spring.

(244) The arm comprises: an upper arm connected to the housing; and a lower arm connected to the mount, wherein the lower arm is movable along the upper arm to change a length of the arm.

(245) The upper arm and the lower arm are telescopically arranged.

(246) The upper arm extends within the lower arm.

(247) The arm includes an arm tensioner configured to urge the arm towards the contracted state, wherein the arm tensioner is disposed within both of the upper arm and the lower arm.

(248) The arm is connected to the housing at a first pivot.

(249) The arm is connected to the mount at a second pivot.

(250) The power tool appliance does not include any gear as part of the drive for translating the rotational motion received from the power tool.

(251) A pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive,

- (252) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump.
- (253) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.
- (254) The pump includes a piston.
- (255) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a drive axis.
- (256) The drive axis is disposed parallel to the pump axis.
- (257) The pump axis is radially offset from the drive axis.
- (258) The pump axis is disposed vertically below the drive axis.
- (259) The pump axis and the drive axis are disposed co-planar on a plane that extends vertically upward and downward relative to the housing and longitudinally forward and rearward relative to the housing.
- (260) The drive includes an overdrive gear configured to receive the rotational motion from the power tool at a first speed and configured to output rotational driving motion at a second speed greater than the first speed.
- (261) A pump supported by the housing, the pump connected to the drive to receive the rotational driving motion from the drive
- (262) The pump is a turbine pump.
- (263) An input shaft configured to connect to the power tool to receive the rotational output from the power tool is not disposed coaxially with the drive.
- (264) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a mount connected to the housing, the mount configured to interface with the power tool to statically connect the power tool to the housing; and a shaft aligner supported on the housing, the shaft aligner configured to bias the housing vertically upwards relative to the power tool.
- (265) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:
- (266) The shaft aligner is configured to exert a variable biasing force.
- (267) The shaft aligner includes at least one spring.
- (268) The shaft aligner is disposed longitudinally between a center of gravity of the power tool appliance and a portion of an input shaft of the power tool appliance that projects out of a rear side of the housing, the input shaft configured to receive the rotational input from the power tool.
- (269) The shaft aligner includes a strut.
- (270) The strut is configured to rock on the housing.
- (271) The shaft aligner includes a first spring and a strut, wherein the first spring is disposed between the strut and the housing such that the first spring is configured to be compressed between the strut and the housing to bias the strut in a first rotational direction about a strut pivot.
- (272) The shaft aligner includes a second spring disposed between the strut and the housing such that the second spring is configured to be compressed between the strut and the housing to bias the strut in the first rotational direction.
- (273) The shaft aligner includes a third spring, the third spring disposed between the strut and the housing such that the third spring is configured to be compressed between the strut and the housing to bias the strut in a second rotational direction opposite the first rotational direction.
- (274) The shaft aligner includes a fourth spring disposed between the strut and the housing such

that the fourth spring is configured to be compressed between the strut and the housing to bias the strut in the second rotational direction.

(275) An arm extending between and connecting the housing and the mount, wherein the strut is disposed between the arm and the housing.

(276) The arm is connected to the housing at on an opposite side of the strut from a front end of the housing.

(277) The arm retains the strut on the housing.

(278) The arm is pivotably connected to the housing.

(279) The mount is pivotably connected to the arm.

(280) The arm extends over a first lateral side of the housing and a second lateral side of the housing, and wherein the strut extends over the first lateral side and the second lateral side.

(281) The arm is movable between a contracted state and an extended state to change a length of the arm.

(282) A tensioner configured to urge the arm towards the contracted state.

(283) The tensioner is disposed within the arm.

(284) The arm includes an upper arm extension and a lower arm extension that are movable relative to each other.

(285) The upper arm extension and the lower arm extension are disposed telescopically.

(286) The upper arm extension is at least partially disposed within the lower arm extension.

(287) A tensioner configured to urge the arm towards the contracted state, the tensioner disposed within and connected to the upper arm extension and the lower arm extension.

(288) A pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive,

(289) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump.

(290) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.

(291) The pump includes a piston.

(292) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a drive axis.

(293) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a support assembly connected to the housing and extending from the housing, the support assembly configured to interface with the power tool to statically connect the power tool to the housing; and a spring disposed between the housing and the support assembly, the spring configured to bias the housing upwards and away from the support assembly to align the power tool and the power tool assembly.

(294) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(295) The spring is disposed between the housing and a strut, spring configured to bias the housing away from the strut.

(296) The strut is configured to engage the support assembly.

(297) The support assembly comprises an arm, the arm connected to the housing and movable relative to the housing.

(298) The spring is configured to bias the housing vertically upward and away from the arm.

(299) The arm is pivotably connected to the housing.

(300) The support assembly includes a mount connected to the housing, wherein the mount is configured to interface with the power tool to statically connect the power tool to the power tool appliance.

(301) The spring is a first spring; and a second spring is disposed between the housing and the support assembly, the second spring configured to bias the housing upwards and away from the support assembly to align the power tool and the power tool assembly.

(302) The first spring is configured to bias the housing in a first vertical direction and the second spring is configured to bias the housing in a second vertical direction opposite the first vertical direction.

(303) The first spring is configured to bias the housing in a first vertical direction and the second spring is configured to bias the housing in the first vertical direction.

(304) The spring is disposed longitudinally between the support assembly and a center of gravity of the power tool appliance.

(305) A pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive,

(306) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump.

(307) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.

(308) The pump includes a piston.

(309) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a drive axis.

(310) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a support assembly connected to the housing and extending from the housing, the support assembly configured to interface with the power tool to statically connect the power tool to the housing; and a first plurality of springs configured to bias the housing upwards and away from the support assembly to align the power tool and the power tool assembly.

(311) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(312) The first plurality of springs includes a first pair of springs.

(313) The first plurality of springs are configured to exert a biasing force longitudinally forward and vertically in a first vertical direction.

(314) A second plurality of springs configured to bias the housing upwards and away from the support assembly to align the power tool and the power tool assembly.

(315) The first plurality of springs are configured to exert a biasing force longitudinally forward and vertically in a second vertical direction.

(316) The second plurality of springs includes a second pair of springs.

(317) A pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive,

(318) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump.

(319) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.

(320) The pump includes a piston.

(321) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive

to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a drive axis.

(322) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a support assembly connected to the housing and extending from the housing, the support assembly configured to interface with the power tool to statically connect the power tool to the housing; a first spring configured to bias the support assembly longitudinally rearward away from the housing and vertically upwards to align the power tool and the power tool assembly; and a tensioner configured to bias a mount of the support assembly to align the power tool and the power tool assembly.

(323) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(324) The tensioner is configured to bias the mount longitudinally forward and vertically upwards relative to the housing.

(325) A pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive,

(326) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump.

(327) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.

(328) The pump includes a piston.

(329) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a drive axis.

(330) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; wherein the power tool appliance is configured to preload the power tool.

(331) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(332) Preloading inhibits the power tool from triggering an impact driver function.

(333) The power tool appliance preloads the power tool through a drive shaft of the power tool appliance.

(334) The power tool appliance preloads the power tool through the drive shaft via one or more springs of the power tool appliance indirectly orientating the power tool relative to the drive shaft.

(335) An arm spring of the one or more springs tensions an arm connected to a mount on which the power tool is mountable.

(336) The arm is extendable by overcoming the arm spring, and the arm spring urging the arm to be shorter.

(337) A pivot member orientates the arm.

(338) The pivot member is urged by at least one of the one or more springs.

(339) The pivot member is a strut.

(340) The strut has an upper side and a lower side, the lower side engaging the arm in a first orientation of the power tool appliance and the upper side engaging the arm in a second orientation

of the power tool appliance that is inverted with respect to the first orientation.

(341) A pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive,

(342) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump.

(343) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.

(344) The pump includes a piston.

(345) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a drive axis.

(346) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; an arm extending from the housing, the arm configured to connect the power tool to the housing; and a strut configured to engage the arm to orient the arm relative to the housing.

(347) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(348) The strut is configured to pivot relative to the housing.

(349) The strut is not directly connected to the housing.

(350) The strut is held on the housing by the arm.

(351) The strut is configured to pivot relative to the housing of the power tool appliance to adjust alignment of the power tool and the power tool appliance via orientation of the arm relative to the housing.

(352) The strut comprises an upper face and a lower face, the strut rocking to alternately extend or withdraw the upper face and the lower face to orientate the arm.

(353) One or more springs which urge the strut.

(354) The one or more springs comprise one or more upper springs and one or more lower springs, the one or more upper springs urging in opposition to the one or more lower springs to balance the strut.

(355) The arm is connected an arm mount to connect to the housing, wherein the arm mount projects through an aperture in the strut.

(356) The arm mount is formed as a portion of the housing.

(357) The arm is mounted to the arm mount at an arm pivot.

(358) The housing includes at least one ridge, the at least one ridge configured to limit displacement of the strut.

(359) The at least one ridge is sloped transverse to a horizontal plane.

(360) The at least one ridge includes a first ridge configured to limit displacement of the strut in a first direction and a second ridge configured to limit displacement of the strut in a second opposite direction.

(361) The first ridge and the second ridge are disposed as a chevron pointing longitudinally rearward.

(362) An input shaft projects from a rear side of the housing, wherein the strut is disposed longitudinally between the input shaft and a front side of the housing.

(363) A pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive,

(364) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet

is configured to receive spray fluid into the pump.

(365) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.

(366) The pump includes a piston.

(367) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a drive axis.

(368) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; and an arm connected to the housing and configured to statically connect the power tool to the housing.

(369) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(370) The arm is movable between a contracted state and an extended state to change a length of the arm.

(371) A tensioner configured to urge the arm to the contracted state.

(372) The tensioner is a spring.

(373) The tensioner is internal to the arm.

(374) The arm includes an upper arm extension and a lower arm extension, and wherein the tensioner is connected to the upper arm extension and the lower arm extension to pull the upper arm extension and the lower arm extension together.

(375) The upper arm extension and lower arm extension are disposed telescopically.

(376) The upper arm extension is at least partially disposed within the lower arm extension.

(377) The arm is movable relative to the housing.

(378) The arm is pivotable relative to the housing.

(379) A pump inlet of a pump supported by the housing projects from a reservoir-facing side of the housing, and wherein the housing is reorientable relative to the arm between a suction feed state, in which the pump inlet is oriented vertically downward, and a gravity feed state, in which the pump inlet is oriented vertically upward.

(380) The arm is connected to the housing by an arm yoke.

(381) An input shaft extends outward from a rear side of the housing, the input shaft disposed laterally between branches of the arm yoke.

(382) The arm comprises: an upper arm extension connected to the housing; and a lower arm extension projecting from the upper arm extension and movable relative to the upper arm extension to adjust a length of the arm.

(383) The upper arm extension includes an arm yoke connected to the housing.

(384) The lower arm extension includes a mount yoke configured to connect to a mount that connects to the power tool.

(385) A pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive,

(386) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump.

(387) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.

(388) The pump includes a piston.

(389) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a

drive axis.

(390) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; and an arm connected to the housing and configured to statically connect the power tool to the housing, wherein the arm is reconfigurable between a contracted state and an extended state to vary a length of the arm.

(391) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(392) A tensioner configured to bias the arm towards the contracted state.

(393) The arm is movable relative to the housing such that the arm is reorientable relative to the housing.

(394) The arm is pivotably connected to the housing.

(395) A mount connected to the arm, the mount configured to interface with the power tool to statically connect the power tool to the housing.

(396) The mount is pivotably connected to the arm.

(397) A pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive,

(398) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump.

(399) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.

(400) The pump includes a piston.

(401) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a drive axis.

(402) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; an input shaft extending out a rear end of the housing, the input shaft configured to interface with the power tool to receive the rotational input; and an arm movably connected to the housing at an arm pivot, the arm configured to statically connect the power tool to the housing.

(403) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(404) The arm is connected to an arm mount to connect to the housing.

(405) The arm mount is formed as a portion of the housing.

(406) A spring configured to bias the housing away from the arm to align the power tool and the power tool appliance.

(407) A strut is disposed between the spring and the arm such that the spring exerts force the arm via the strut.

(408) The arm mount extends through the strut.

(409) The strut is disposed around a strut mount, the arm mount extending longitudinally rearward from the strut mount.

(410) The strut mount is radially larger than the arm mount relative to a drive axis of the drive.

(411) The strut is configured to pivot relative to the housing.

- (412) The strut is configured to rock on a strut pivot of the housing.
- (413) The strut pivot is disposed longitudinally between a center of gravity of the power tool appliance and the arm pivot.
- (414) The strut pivot is disposed longitudinally between a pump inlet of a pump and the arm pivot, the pump supported by the housing and connected to the drive to be powered by the drive.
- (415) The arm includes an arm yoke connected to the arm pivot.
- (416) The arm yoke includes a first branch disposed on a first lateral side of the housing and a second branch disposed on a second lateral side of the housing.
- (417) A pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive,
- (418) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump.
- (419) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.
- (420) The pump includes a piston.
- (421) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a drive axis.
- (422) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive including a drive shaft configured to rotate on a drive axis; a rotational input shaft projecting rearward relative to a rear side of the housing, the rotational input shaft configured to dynamically connect the drive to the power tool to provide the rotational input from the power tool to the drive, wherein the rotational input shaft is unitary with the drive shaft.
- (423) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:
- (424) The rotational input shaft includes a keyed exterior.
- (425) The keyed exterior is hexed.
- (426) The rotational input shaft includes a shaft groove.
- (427) The keyed exterior extends along the rotational input shaft in a first axial direction relative to the shaft groove and in a second axial direction relative to the shaft groove.
- (428) The rotational input shaft includes a head and a shaft body, the shaft groove disposed axially between the head and the shaft body.
- (429) The keyed exterior is formed on the shaft body and the head.
- (430) The drive includes: a drive body that is radially enlarged relative to the rotational input shaft, the drive body monolithic with the rotational input shaft; an inner groove formed on the drive body, the inner groove extending about and canted relative to the drive axis; an outer ring disposed about the drive body, the outer ring including an output configured to cause pumping by the pump; an outer groove formed on the outer ring; a plurality of balls disposed in the inner groove and the outer groove; wherein rotation of the drive body causes the outer ring to rock to linearly reciprocate the output.
- (431) The drive is rotatably supported by a first bushing and a second bushing, the drive body disposed axially between the first bushing and the second bushing.
- (432) The rotational input shaft is cantilevered from the second bushing.
- (433) The rotational input shaft and the drive body are formed from a single piece of metal.
- (434) A pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive,
- (435) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet

is configured to receive spray fluid into the pump.

(436) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.

(437) The pump includes a piston.

(438) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a drive axis.

(439) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive including a drive shaft configured to rotate on a drive axis; a rotational input shaft projecting rearward relative to a rear side of the housing, the rotational input shaft configured to dynamically connect the drive to the power tool to provide the rotational input from the power tool to the drive, wherein the rotational input shaft includes a keyed exterior.

(440) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(441) The keyed exterior is hexed.

(442) The input shaft is part of a single piece of metal that also converts, at least in part, rotational motion to reciprocating motion.

(443) A pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive,

(444) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump.

(445) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.

(446) The pump includes a piston.

(447) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a drive axis.

(448) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive including a drive shaft configured to rotate on a drive axis; a rotational input shaft projecting rearward relative to a rear side of the housing, the rotational input shaft configured to dynamically connect the drive to the power tool to provide the rotational input from the power tool to the drive, wherein the rotational input shaft and a portion of the drive configured to exert axial force to cause axial movement of an output of the drive are formed from a single piece of metal.

(449) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(450) The rotational input shaft includes a keyed exterior.

(451) The keyed exterior is hexed.

(452) A pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive,

(453) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump.

(454) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.

(455) The pump includes a piston.

(456) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a drive axis.

(457) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive including a drive shaft configured to rotate on a drive axis; a rotational input shaft projecting rearward relative to a rear side of the housing, the rotational input shaft configured to dynamically connect the drive to the power tool to provide the rotational input from the power tool to the drive, wherein the rotational input shaft is anchored in the housing and is connectable to the power tool to receive the rotational input, wherein the rotational input shaft is not part of the power tool.

(458) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(459) A pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive,

(460) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump.

(461) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.

(462) The pump includes a piston.

(463) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a drive axis.

(464) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive, wherein the pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump; and a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray; wherein the power tool appliance is reconfigurable between a suction feed state in which the pump is configured to draw the spray fluid upward into the pump and a gravity feed state in which the spray fluid is configured to flow downward into the pump.

(465) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(466) The pump inlet is oriented vertically downward with the power tool appliance in the suction feed state and the pump inlet is oriented vertically upward with the power tool appliance in the gravity feed state.

(467) The housing is configured to be inverted between the suction feed state in which the pump inlet faces downward and the gravity feed state in which the pump inlet faces upward.

(468) A connector configured to interface with a first reservoir to connect the first reservoir to the pump and configured to interface with a second reservoir to connect the second reservoir to the pump.

(469) The connector is configured to connect to the first reservoir such that the first reservoir hangs

down below the housing.

(470) The connector is configured to connect to the second reservoir such that the second reservoir extends above the connector to hold the spray fluid above the pump inlet.

(471) The connector includes a plurality of tabs.

(472) The connector is formed by the pump body.

(473) An arm connected to and extending from the housing, the arm configured to statically connect the power tool and the housing to prevent relative rotation therebetween; wherein the arm is movable relative to the housing such that the housing can be repositioned relative to the arm between the suction feed state and the gravity feed state.

(474) The arm is pivotably connected to the housing.

(475) The arm includes an arm yoke that is pivotably connected to the housing.

(476) The arm yoke extends over lateral sides of the housing to connect to the housing.

(477) A portion of the housing is disposed between a first branch of the arm yoke and a second branch of the arm yoke, the first branch and the second branch pivotably connected to the housing.

(478) The housing is connected to the arm such that an input shaft that extends rearward relative to a rear side of the housing moves through the arm yoke during reconfiguration between the suction feed state and the gravity feed state, wherein the input shaft is configured to be received by the power tool to receive the rotational input.

(479) A front end of the housing does not move through the arm yoke during reconfiguration between the suction feed state and the gravity feed state.

(480) The pump inlet does not move through the arm yoke during reconfiguration between the suction feed state and the gravity feed state.

(481) A mount connected to the arm such that the mount is connected to the housing by the arm, the mount configured to interface with the power tool.

(482) The mount is movable relative to the arm.

(483) The mount is pivotable relative to the arm.

(484) The arm is configured to pivot relative to the housing on a first pivot axis, the mount is configured to pivot relative to the arm on a second pivot axis, and the first pivot axis is parallel to the second pivot axis.

(485) The mount includes a first interface side and a second interface side, the first interface side configured to be oriented face up to contact the power tool with the power tool appliance in the suction feed state, and the second interface side configured to be oriented face up to contact the power tool with the power tool appliance in the gravity feed state.

(486) A slot extends laterally through the mount, the slot configured such that a fixation is capable of being passed through the mount to wrap around a base of the power tool to fix the power tool to the mount.

(487) The mount is formed as a plate.

(488) The arm is disposed directly longitudinally forward of a handle of the power tool with the power tool appliance in both the suction feed state and the gravity feed state such that the arm forms at least a portion of a trigger guard for the power tool.

(489) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive, wherein the pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump; a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray; and an arm connected to the housing and movable relative to the housing, the arm configured to statically connect the power tool to the

housing; wherein the housing is configured to pivot relative to the arm between a suction feed state, in which the pump inlet is oriented downward such that the pump is configured to draw the spray fluid upward into the pump, and a gravity feed state, in which the pump inlet is oriented upward such that the spray fluid is configured to flow downward into the pump.

(490) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(491) The arm is pivotably connected to the housing.

(492) A rotational input shaft supported in the housing is disposed rearward of a rear end of the housing, the spray orifice oriented to emit the fluid spray forward relative to a forward end of the housing.

(493) The rotational input shaft is configured to pass directly between a first branch and a second branch of an arm yoke of the arm that is pivotably connected to the housing as the housing is flipped between the suction feed state and the gravity feed state.

(494) The pump inlet does not pass directly between a first branch and a second branch of an arm yoke of the arm that is pivotably connected to the housing as the housing is flipped between the suction feed state and the gravity feed state.

(495) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive, wherein the pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump; and a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray; wherein the housing is configured to flip between a suction feed state, in which the pump inlet is oriented downward such that the pump is configured to draw the spray fluid upward into the pump, and a gravity feed state, in which the pump inlet is oriented upward such that the spray fluid is configured to flow downward into the pump.

(496) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive, wherein the pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump; a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray; an arm connected to the housing and movable relative to the housing; and a mount connected to the arm and movable relative to the arm; wherein the mount and the arm are configured to statically connect the power tool to the housing; and wherein the housing is configured to pivot relative to the arm between a suction feed state, in which the pump inlet is oriented downward such that the pump is configured to draw the spray fluid upward into the pump, and a gravity feed state, in which the pump inlet is oriented upward such that the spray fluid is configured to flow downward into the pump.

(497) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(498) The mount includes a first interface side and a second interface side, the first interface side configured to be oriented face up to contact the power tool with the housing disposed in the suction

feed state, and the second interface side configured to be oriented face up to contact the power tool with the housing disposed in the gravity feed state.

(499) The mount is configured to pivot relative to the arm to reoriented between the first interface side being oriented face up and the second interface side being oriented face up.

(500) The housing is configured to reoriented relative to the arm such that: a first side of the arm is oriented longitudinally forward relative to the housing with the housing disposed in the suction feed state; and the first side of the arm is oriented longitudinally rearward relative to the housing with the housing disposed in the gravity feed state.

(501) The arm is disposed longitudinally forward of a handle of the power tool with the housing in both the suction feed state and the gravity feed state.

(502) The arm is disposed longitudinally between the pump inlet and a handle of the power tool with the housing in both the suction feed state and the gravity feed state.

(503) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a mount connected to the housing, the mount configured to interface with the power tool to statically connect the power tool to the housing; and a fixation configured to wrap around the mount and a base of the power tool to fix the power tool to the mount.

(504) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(505) The fixation is a strap.

(506) The strap includes hook-and-loop fasteners.

(507) The strap includes double-sided hook and loop fasteners such that a first side of the strap includes both hooks and loops and a second side of the strap includes both hooks and loops.

(508) The fixation includes double-sided hook-and-loop fasteners such that the fixation is configured to remain connected to the mount during flipping of the power tool assembly between a suction feed state and a gravity feed state.

(509) At least one slot extends through the mount, the fixation extending through the at least one slot.

(510) The at least one slot includes a plurality of slots.

(511) The mount includes a rib disposed between a first slot of the plurality of slots and a second slot of the plurality of slots, the rib extending between and connecting a first interface side of the mount and a second interface side of the mount.

(512) The mount comprises: a first interface side configured to be oriented face up to support the power tool with the housing in a suction feed state in which the pump inlet is oriented downward; a second interface side configured to be oriented face up to support the power tool with the housing in a gravity feed state in which the pump inlet is oriented upward; and a slot open laterally through the mount and disposed between the first interface side and the second interface side, the slot configured such that the fixation passes through the slot to fix the power tool to the mount.

(513) The mount further comprises: a laterally elongate rib extending between and connecting the first interface side and the second interface side, the laterally elongate rib at least partially defining the slot and at least partially defining a second slot extending laterally through the mount.

(514) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(515) The rotational input shaft includes a keyed exterior.

(516) The keyed exterior is hexed.

(517) A pump at least partially disposed within the housing, the pump connected to the drive to

receive an output from the drive,

(518) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump.

(519) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.

(520) The pump includes a piston.

(521) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a drive axis.

(522) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a mount connected to the housing, the mount configured to interface with the power tool to statically connect the power tool to the housing; and a fixation configured to extend through a slot that extends laterally through the mount to wrap around the mount and a base of the power tool to fix the power tool to the mount.

(523) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(524) The fixation is a strap.

(525) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(526) The rotational input shaft includes a keyed exterior.

(527) The keyed exterior is hexed.

(528) A pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive,

(529) The pump includes a pump inlet that is exposed through the housing, wherein the pump inlet is configured to receive spray fluid into the pump.

(530) A spray outlet disposed downstream of the pump, the spray outlet configured to atomize output from the pump into a fluid spray.

(531) The pump includes a piston.

(532) The piston is configured to reciprocate along a pump axis, the piston is connected to the drive to receive a reciprocating linear output from the drive, and the drive is configured to rotate on a drive axis.

(533) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance comprising: a housing; a turbine at least partially disposed within the housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool and configured to provide rotational motion to the turbine to drive the turbine; a duct disposed downstream of the turbine, the duct narrowing from the turbine and configured to direct airflow to a spray orifice; and a mount connected to the housing, the mount configured to interface with the power tool to statically connect the power tool to the housing.

(534) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(535) The drive is configured to receive an input at a first rotational speed and provide an output at a second rotational speed greater than the first rotational speed.

(536) The drive includes an overdrive gear set configured to increase a speed of the rotational input.

(537) A method of assembling a tool system, the method comprising: inserting an input shaft extending out from a housing of a power tool appliance into a bit receiver of a power tool; mounting the power tool to a mount of the power tool appliance to statically connect the power tool relative to the housing; and pulling the trigger of the power tool to cause an output of the power tool appliance.

(538) The method of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(539) Inserting the input shaft is performed before mounting the power tool to the mount.

(540) Mounting the power tool to the mount includes wrapping a strap around a base of the power tool.

(541) Mounting the power tool to the mount includes securing a battery of the power tool onto a plate of the power tool appliance.

(542) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance including a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive, wherein the pump includes an inlet that is exposed through the housing, wherein the inlet is configured to receive spray fluid into the pump; a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray; a support assembly connected to the housing and extending from the housing, the support assembly configured to interface with the power tool to statically connect the power tool to the housing; and a biaser configured to bias the support assembly to align the power tool and the drive.

(543) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(544) The biaser is a spring disposed between the housing and the support assembly, the biaser configured to bias the housing relative to the support assembly.

(545) The biaser is a spring configured to set an angle between the support assembly and the housing.

(546) The biaser is a spring configured to set a length of the support assembly.

(547) The spring is disposed within the support assembly.

(548) The spring is configured to bias the support assembly towards a contracted state.

(549) The biaser is a spring is disposed between the housing and a strut, spring configured to bias the housing away from the strut.

(550) The strut is configured to engage the support assembly.

(551) The support assembly comprises an arm, the arm connected to the housing and movable relative to the housing.

(552) The biaser is a spring configured to set an angle of the arm relative to the housing.

(553) The biaser is a spring is configured to set a length of the arm.

(554) The biaser is a spring is configured to bias the housing vertically upward and away from the arm.

(555) The arm is pivotably connected to the housing.

(556) The support assembly includes a mount connected to the housing by the arm, wherein the mount is configured to interface with the power tool to statically connect the power tool to the power tool appliance.

(557) The support assembly comprises an arm, wherein the arm is adjustable to change a length of the arm, and wherein the biaser is configured to bias the arm into a contracted state.

(558) The arm includes an upper arm extension connected to and extending from the housing, and a lower arm extension extending from the upper arm extension, wherein the upper arm extension and the lower arm extension are telescopically disposed.

(559) The biaser is a first spring disposed between the housing and the support assembly; a second spring is disposed between the housing and the support assembly; and the first spring and the second spring are configured to bias the housing upwards and away from the support assembly to align the power tool and the power tool assembly.

(560) The first spring is configured to bias the housing in a first vertical direction and the second spring is configured to bias the housing in a second vertical direction opposite the first vertical direction.

(561) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance including a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive, wherein the pump includes an inlet that is exposed through the housing, wherein the inlet is configured to receive spray fluid into the pump; a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray; a support assembly including an arm connected to the housing and extending from the housing, the support assembly configured to interface with the power tool to statically connect the power tool to the housing; and a spring disposed between the arm and the housing, the spring configured to bias the housing upward and away from the support assembly to align the power tool and the drive.

(562) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance including a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool; a pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive, wherein the pump includes an inlet that is exposed through the housing, wherein the inlet is configured to receive spray fluid into the pump; a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray; a support assembly including an arm connected to the housing and extending from the housing, the support assembly configured to interface with the power tool to statically connect the power tool to the housing; and a spring disposed within the arm, the spring configured to adjust a length of the arm and configured to bias the arm towards a contracted state to align the power tool and the drive.

(563) A power tool appliance configured for use with a power tool that outputs rotational motion, the power tool appliance configured to attach to and receive rotational input from the power tool, the power tool appliance including a housing; a drive at least partially disposed within the housing, the drive configured to dynamically connect to the power tool to receive the rotational input from the power tool, the drive configured to rotate on a drive axis; a pump at least partially disposed within the housing, the pump connected to the drive to receive an output from the drive, wherein the pump includes an inlet that is exposed through the housing, wherein the inlet is configured to receive spray fluid into the pump; a spray outlet disposed downstream of the pump, the spray outlet configured to atomize the spray fluid output by the pump into a fluid spray; and a support assembly connected to the housing and extending from the housing, the support assembly configured to interface with the power tool to statically connect the power tool to the housing; wherein a piston of the pump is vertically aligned with the drive and radially offset from the drive axis.

(564) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(565) The piston is configured to reciprocate on a pump axis, and wherein the pump axis is disposed parallel to the drive axis.

(566) The piston is configured to reciprocate on a pump axis, wherein the pump axis and the drive axis are co-planar on a plane that extends longitudinally and vertically.

(567) A power tool appliance for use with a power tool that outputs rotational motion for one or both of drilling and fastening, the power tool appliance configured to attached to, and receive rotational input from, the power tool, the power tool appliance including a housing; a drive at least partially located within the housing, the drive comprising an input shaft having an end that is configured to be received by the power tool, the input shaft extending into the housing, the drive converting rotational motion received from the power tool through the input shaft into a reciprocating motion; and a support connected to the housing, the support configured to interface with the power tool to statically connect the power tool to the housing, the support comprising an arm that extends away from the housing and that connects between the power tool and the housing.

(568) The power tool appliance of the preceding paragraph can optionally include, additionally and/or alternatively, any one or more of the following features, configurations and/or additional components:

(569) The support comprises a strap that wraps at least partially around a battery portion of the power tool to brace the power tool relative to the housing.

(570) The strap includes hook and loop fasteners.

(571) The support comprises a plate on which the battery portion of the power tool stands, and wherein the strap extends through or around the plate such that the strap holds the battery portion of the power tool against the plate.

(572) The arm pivots relative to the housing.

(573) The end of the input shaft is multi-faceted.

(574) Material that forms the input shaft is contiguous with material that forms at least part of an eccentric of the drive.

(575) The arm is biased by at least one spring to urge the power tool relative to the housing.

(576) The at least one spring urges the arm to shorten which, due to the power tool being connected to the housing via the arm of the support, urges the power tool relative to the housing.

(577) One or more spring of the at least one spring is located inside of the arm, the arm comprises telescoping parts to shorten and lengthen the arm, and the one or more spring of the at least one spring urges the arm to shorten or lengthen.

(578) One or more spring of the at least one spring is located outside of the arm and urges a strut of the power tool appliance.

(579) The strut engages the arm which, due to the power tool being connected to the housing via the arm of the mount, urges the power tool relative to the housing.

(580) The one or more spring of the at least one spring is located between the housing and the strut such that the one or more spring pushes the strut away from the housing.

(581) The at least one spring urges the power tool to pre-load the power tool to minimize entry into a mode of the power tool.

(582) The at least one spring urges the power tool to align relative to the housing of the power tool appliance.

(583) The support comprises a plate on which the battery portion of the power tool stands.

(584) The plate is connected to the arm by a hinge which permits the plate to pivot relative to the arm.

(585) A pump that receives the reciprocating motion from the drive which operates the pump.

(586) A spray nozzle that sprays fluid output by the pump and a reservoir supported by the housing,

the pump in fluid communication with the reservoir.

(587) A method of using a power tool appliance to convert rotational motion output from a power tool to reciprocating motion, the method including mounting the power tool to the power tool appliance, the power tool appliance comprising: a housing; a drive at least partially located within the housing, the drive comprising an input shaft, the input shaft extending out from the housing; and a support connected to the housing, the support configured to interface with the power tool to statically connect the power tool to the housing, the support comprising an arm that extends away from the housing and which connects between the power tool and the housing, wherein mounting comprises the input shaft being received within the power tool; and operating the power tool appliance via the power tool by the drive converting rotational motion received from the power tool through the input shaft into a reciprocating motion.

(588) While the invention has been described with reference to an exemplary embodiment(s), it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment(s) disclosed, but that the invention will include all embodiments falling within the scope of the appended claims.

Claims

1. A power tool appliance for use with a power tool that outputs rotational motion for one or both of drilling and fastening, the power tool appliance configured to attached to, and receive rotational input from, the power tool, the power tool appliance comprising: a housing; a drive at least partially located within the housing, the drive comprising an input shaft having an end that is configured to be received by the power tool, the drive converting rotational motion received from the power tool through the input shaft into a reciprocating motion; and a support connected to the housing, the support configured to interface with the power tool to statically connect the power tool to the housing, the support comprising an arm that extends away from the housing and that connects between the power tool and the housing; wherein the support comprises a plate on which a battery portion of the power tool stands and a strap that extends around at least part of the plate, wherein the strap wraps at least partially around the battery portion of the power tool to brace the power tool relative to the housing and hold the battery portion of the power tool against the plate.
2. The power tool appliance of claim 1, wherein the strap includes hook and loop fasteners.
3. The power tool appliance of claim 2, wherein the arm pivots relative to the housing.
4. The power tool appliance of claim 1, wherein the end of the input shaft is multi-faceted.
5. The power tool appliance of claim 1, wherein material that forms the input shaft is contiguous with material that forms at least part of an eccentric of the drive.
6. A power tool appliance for use with a power tool that outputs rotational motion for one or both of drilling and fastening, the power tool appliance configured to attached to, and receive rotational input from, the power tool, the power tool appliance comprising: a housing; a drive at least partially located within the housing, the drive comprising an input shaft having an end that is configured to be received by the power tool, the drive converting rotational motion received from the power tool through the input shaft into a reciprocating motion; and a support connected to the housing, the support configured to interface with the power tool to statically connect the power tool to the housing, the support comprising an arm that extends away from the housing and that connects between the power tool and the housing; wherein the arm is biased by at least one spring to urge the power tool relative to the housing.
7. The power tool appliance of claim 6, wherein the at least one spring urges the arm to shorten which, due to the power tool being connected to the housing via the arm of the support, urges the

power tool relative to the housing.

8. The power tool appliance of claim 7, wherein one or more spring of the at least one spring is located inside of the arm, the arm comprises telescoping parts to shorten and lengthen the arm, and the one or more spring of the at least one spring urges the arm to shorten or lengthen.

9. The power tool appliance of claim 6, wherein one or more spring of the at least one spring is located outside of the arm and urges a strut of the power tool appliance.

10. The power tool appliance of claim 9, wherein the strut engages the arm which, due to the power tool being connected to the housing via the arm of the support, urges the power tool relative to the housing.

11. The power tool appliance of claim 9, wherein the one or more spring of the at least one spring is located between the housing and the strut such that the one or more spring pushes the strut away from the housing.

12. The power tool appliance of claim 6, wherein the at least one spring urges the power tool to pre-load the power tool to minimize entry into a mode of the power tool.

13. The power tool appliance of claim 6, wherein the at least one spring urges the power tool to align relative to the housing of the power tool appliance.

14. A power tool appliance for use with a power tool that outputs rotational motion for one or both of drilling and fastening, the power tool appliance configured to attached to, and receive rotational input from, the power tool, the power tool appliance comprising: a housing; a drive at least partially located within the housing, the drive comprising an input shaft having an end that is configured to be received by the power tool, the drive converting rotational motion received from the power tool through the input shaft into a reciprocating motion; and a support connected to the housing, the support configured to interface with the power tool to statically connect the power tool to the housing, the support comprising an arm that extends away from the housing and that connects between the power tool and the housing; wherein the support comprises a plate on which the battery portion of the power tool stands, and wherein the plate is connected to the arm by a hinge which permits the plate to pivot relative to the arm.

15. The power tool appliance of claim 14, further comprising a pump that receives the reciprocating motion from the drive which operates the pump.

16. The power tool appliance of claim 15, further comprising a spray nozzle that sprays fluid output by the pump and a reservoir supported by the housing, the pump in fluid communication with the reservoir.
